1. If $z + z^3 = 0$ then which of the following must be true on the complex plane?

यदि $z + z^3 = 0$ तब सम्मिश्र समतल में निम्न में से कौनसा निश्चित ही सत्य होगा :

(A) $Re(z) < 0$ (B) $Re(z) = 0$ (C) $Im(z) = 0$ (D) $z^4 = 1$

Ans. B

Sol. $z(1 + z^2) = 0 \Rightarrow z = 0$ or $z^2 = i^2 \Rightarrow z = 0$ or $z = \pm i \Rightarrow Re(z) = 0$

2. Number of integral values of n for which the quantity $(n + i)^4$, where $i^2 = -1$, is an integer is :

n के पूर्णांक मानों की संख्या, यदि $(n + i)^4$ एक पूर्णांक है : (जहां $i^2 = -1$)

(A) 1 (B) 2 (C) 3 (D) 4

Ans. C

Sol. $(n + i)^4 = n^4 + 4n^3i + 6n^2i^2 + 4ni^3 + i^4$

$= n^4 - 6n^2 + 1 + i(4n^3 - 4n)$

for this to be integer

$4n^3 - 4n = 4n(n^2 - 1)$ must be zero

$\Rightarrow$ $n = 0$ or $n = \pm 1 \Rightarrow$ 3 values $\Rightarrow$ (C)

3. Let $i = \sqrt{-1}$. The product of the real part of the roots of $z^2 - z = 5 - 5i$ is :

$z^2 - z = 5 - 5i$ के मूलों के वास्तविक भाग का गुणनफल हैं : (जहां $i = \sqrt{-1}$)

(A) – 25 (B) – 6 (C) – 5 (D) 25

Ans. B

Sol. roots are $3 - i$ and $-2 + i \Rightarrow -6$

4. There is only one way to choose real numbers M and N such that when the polynomial $5x^4 + 4x^3 + 3x^2 + Mx + N$ is divided by the polynomial $x^2 + 1$, the remainder is 0. If M and N assume these unique values, then M – N is :

वास्तविक संख्याओं M और N को चुनने का केवल एक तरीका इस प्रकार है कि $5x^4 + 4x^3 + 3x^2 + Mx + N$, बहुपद $x^2 + 1$ से विभाज्य है, तब M – N =

(A) – 6 (B) – 2 (C) 6 (D) 2

Ans. C

Sol. Let $P(x) = 5x^4 + 4x^3 + 3x^2 + Mx + N$

let $Q(x) = x^2 + 1$

if the quotient is Q

then $P(x) = Q(x^2 + 1)$

if $x = i$ then $P(i) = 0$

if $x = -i$ then $P(-i) = 0$

hence $5 - 4i - 3 + Mi + N = 0$

hence $N + Mi = -2 + 4i$

$\therefore$ $N = -2$; $M = 4$

$\therefore$ $M - N = 6$

5. In the quadratic equation $x^2 + (p + iq)x + 3i = 0$, p & q are real. If the sum of the squares of the roots is 8 then :

द्विघात समीकरण $x^2 + (p + iq)x + 3i = 0$ में p और q वास्तविक है यदि मूलों के वर्गो का योग 8 है तब :

(A) $p = 3,\ q = -1$ (B) $p = -3, q = -1$ (C) $p = \pm 3,\ q = \pm 1$ (D) $p = -3,\ q = 1$

Ans. C

Sol. $\alpha + \beta = -(p + iq);\ \alpha\beta = 3i$

Given: $\alpha^2 + \beta^2 = 8$

$(\alpha + \beta)^2 - 2\alpha\beta = 8$

$(p + iq)^2 - 6i = 8$

$p^2 - q^2 + i(2pq - 6) = 8 \Rightarrow p^2 - q^2 = 8$ and $pq = 3$

$\Rightarrow$ $p = 3$ & $q = 1$ or $p = -3$ and $q = 1$

6. The complex number z satisfying $z + |z| = 1 + 7i$ then the value of $|z|^2$ equals :

सम्मिश्र संख्या z, समीकरण $z + |z| = 1 + 7i$ को सन्तुष्ट करती है, तब $|z|^2$ बराबर होगा :

(A) 625 (B) 169 (C) 49 (D) 25

Ans. A

Sol. $z = x + iy$

$\therefore$ $x + iy + \sqrt{x^2 + y^2} = 1 + 7i$

$x + \sqrt{x^2 + y^2} = 1$ ... (1)

and $y = 7$ ... (2)

$\therefore$ $x + \sqrt{x^2 + 49} = 1$

$x^2 + 49 = 1 + x^2 - 2x$

$2x = -48$

$x = -24$

$\therefore$ $|z|^2 = x^2 + y^2 = 625$

7. If z and w are complex numbers satisfying $\bar{z} + i\bar{w} = 0$ and Amp(zw) = π, then Amp(z) is equal to :

यदि z और w सम्मिश्र संख्याऐं $\bar{z} + i\bar{w} = 0$ और Amp(zw) = π को सन्तुष्ट करते है, तब Amp(z) बराबर :

(A) $\frac{\pi}{4}$ (B) $\frac{-\pi}{2}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{4}$

Ans. D

Sol. Given $\bar{z} + i\bar{w} = 0 \quad \Rightarrow \quad \bar{z} = -i\bar{w} \quad$ or $\quad z = iw$

$\text{amp}(z) - \text{amp}(w) = \text{amp}\, i = \frac{\pi}{2}$ ....(1)

also $\text{amp}(zw) = \pi$

$\text{amp}(z) + \text{amp}(w) = \pi$ ....(2)

(1) + (2), gives $2\,\text{amp}(z) = \frac{3\pi}{2} \Rightarrow \quad \text{amp}(z) = \frac{3\pi}{4}$; Also $\text{amp}(w) = \frac{\pi}{4}$

8. Number of values of z (real or complex) simultaneously satisfying the system of equations $1 + z + z^2 + z^3 + \ldots\ldots\ldots + z^{17} = 0$ and $1 + z + z^2 + z^3 + \ldots\ldots\ldots + z^{13} = 0$ is :

z के (वास्तविक या काल्पनिक) मानों की संख्या जोकि समीकरणों $1 + z + z^2 + z^3 + \ldots\ldots\ldots + z^{17} = 0$ और $1 + z + z^2 + z^3 + \ldots\ldots\ldots + z^{13} = 0$ को सन्तुष्ट करती है :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. A

Sol. $1 - z^{18} = 0$; $1 - z^{14} = 0 \quad \Rightarrow \quad z^{14} = 1$ or $z^{18} = 1$

since one is extraneous root $z = -1$ is the common root.

9. If $\frac{x-3}{3+i} + \frac{y-3}{3-i} = i$ where $x, y \in R$ then :

(A) $x = 2$ and $y = -8$ (B) $x = -2$ and $y = 8$ (C) $x = -2$ and $y = -6$ (D) $x = 2$ and $y = 8$

यदि $\frac{x-3}{3+i} + \frac{y-3}{3-i} = i$ जहां $x, y \in R$ तब :

(A) $x = 2$ तथा $y = -8$ (B) $x = -2$ तथा $y = 8$ (C) $x = -2$ तथा $y = -6$ (D) $x = 2$ तथा $y = 8$

Ans. B

Sol. $\frac{x-3}{3+i} + \frac{y-3}{3-i} = i$

$\frac{3x - xi - 9 + 3i + 3y + iy - 9 - 3i}{9+1} = i$

$3x + 3y - 18 + i(y - x) = 10i$

$3(x + y - 6) = 0$

$x + y - 6 = 0$

$y - x = 10$

$x = -2, y = 8$

10. If $x = 9^{1/3}\ 9^{1/9}\ 9^{1/27}$ .....ad inf $y = 4^{1/3}\ 4^{-1/9}\ 4^{1/27}$ ..... ad inf and $z = \sum_{r=1}^{\infty} (1 + i)^{-r}$ then, the argument of the complex number $w = x + yz$ is :

यदि $x = 9^{1/3}\ 9^{1/9}\ 9^{1/27}$ .....ad inf $y = 4^{1/3}\ 4^{-1/9}\ 4^{1/27}$ ..... ad inf और $z = \sum_{r=1}^{\infty} (1 + i)^{-r}$ तब $w = x + yz$ का कोणांक है :

(A) 0 (B) $\pi - \tan^{-1}\left(\frac{\sqrt{2}}{3}\right)$ (C) $-\tan^{-1}\left(\frac{\sqrt{2}}{3}\right)$ (D) $-\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$

Ans. C

Sol. $x = 9^{\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+.....} = 9^{\frac{\frac{1}{3}}{1-\frac{1}{3}}} = 9^{\frac{1}{2}} = 3$

$y = 4^{\frac{1}{3}-\frac{1}{9}+\frac{1}{27}+.....} = 4^{\frac{\frac{1}{3}}{1+\frac{1}{3}}} = 4^{\frac{1}{4}} = \sqrt{2}$

$z = \sum_{r=1}^{\infty}(1+i)^{-r} = \frac{1}{1+i} + \frac{1}{(1+i)^2} + \frac{1}{(1+i)^3} + ... = \frac{\frac{1}{1+i}}{1-\frac{1}{1+i}} = \frac{1}{i} = -i$

Let $\omega = x + yz = 3 - \sqrt{2}\ i$ (4th quad.) $\Rightarrow$ $\text{Arg}\ \omega = -\tan^{-1}\left(\frac{\sqrt{2}}{3}\right) \Rightarrow$ (C)

11. Let $z = 9 + bi$ where b is non zero real and $i^2 = -1$. If the imaginary part of z^2 and z^3 are equal, then b^2 equals :

माना $z = 9 + bi$ जहां b अशून्य वास्तविक संख्या है और $i^2 = -1$. यदि z^2 और z^3 के काल्पनिक भाग समान है, तब b^2 बराबर :

(A) 261 (B) 225 (C) 125 (D) 361

Ans. B

Sol. $z^2 = 81 - b^2 + 18bi$

$z^3 = 729 + 243bi - 27b^2 - b^3i$

hence $243b - b^3 = 18b$ and

$243 - b^2 = 18$

$b^2 = 225$

12. The value of the sum $\sum_{n=1}^{13}\left(i^n + i^{n+1}\right)$, where $i = \sqrt{-1}$, equals :

$\sum_{n=1}^{13}\left(i^n + i^{n+1}\right)$ का मान होगा : (जहाँ $i = \sqrt{-1}$)

(A) i (B) $i - 1$ (C) $-i$ (D) 0

Ans. B

Sol. Sum of any four consecutive powers of iota is zero.

$$\therefore \sum_{n=1}^{13}\left(i^n + i^{n+1}\right) = \left(i + i^2 + ... + i^{13}\right) + \left(i^2 + i^3 + ... + i^{14}\right) = i + i^2 = i - 1$$

13. If $z^2 = (\bar{z})^2$, then which statement is true :

(A) z is purely imaginary (B) z is real

(C) $z = -\bar{z}$ (D) z is real or purely imaginary

यदि $z^2 = (\bar{z})^2$ है, तो कौनसा कथन सत्य है :

(A) z शुद्ध काल्पनिक है (B) z वास्तविक है

(C) $z = -\bar{z}$ (D) z वास्तविक या शुद्ध काल्पनिक है

Ans. D

Sol. Let $z = x + iy$, then its coonjucate $\bar{z} = x - iy$

Given that $z^2 = \left(\bar{z}\right)^2$

$\Rightarrow x^2 - y^2 + 2ixy = x^2 - y^2 - 2ixy \qquad \Rightarrow 4ixy = 0$

If $x \neq 0$ then $y = 0$ and if $y \neq 0$

then $x = 0$

14. If $z = x + iy$, then $\frac{z-\bar{z}}{z+\bar{z}}$ is equal to :

यदि $z = x + iy$, तो $\frac{z-\bar{z}}{z+\bar{z}}$ का मान है :

(A) $i(y/x)$ (B) y/x (C) $i(x/y)$ (D) x/y

Ans. A

Sol. $\frac{z-\bar{z}}{z+\bar{z}} = \frac{x+iy-x+iy}{x+iy+x-iy}$

$= \frac{2iy}{2x} = \frac{iy}{x}$

15. Simplify and express the result in the form of $a + bi$:

निम्नलिखित को $a + bi$ के रूप में व्यक्त कीजिये :

(a) $\left(\frac{1+2i}{2+i}\right)^2$ (b) $-i(9 + 6i)(2 - i)^{-1}$

(c) $\left(\frac{4i^3 - i}{2i+1}\right)^2$ (d) $\frac{3+2i}{2-5i} + \frac{3-2i}{2+5i}$

(e) $\frac{(2+i)^2}{2-i} - \frac{(2-i)^2}{2+i}$

(f) A square $P_1P_2P_3P_4$ is drawn in the complex plane with P_1 at (1, 0) and P_3 at (3, 0). Let P_n denotes the point (x_n, y_n) n = 1, 2, 3, 4

Find the numerical value of the product of complex numbers $(x_1 + i\,y_1)(x_2 + i\,y_2)(x_3 + i\,y_3)(x_4 + i\,y_4)$.

(f) सम्मिश्र तल पर एक वर्ग $P_1P_2P_3P_4$ बनाया जाता है जहाँ P_1 (1, 0) तथा P_3 (3, 0) है। माना P_n बिन्दु (x_n, y_n) n = 1, 2, 3, 4 को प्रदर्शित करता है।

सम्मिश्र संख्याओं के गान $(x_1 + i\,y_1)(x_2 + i\,y_2)(x_3 + i\,y_3)(x_4 + i\,y_4)$ का मान ज्ञात कीजिये।

Ans. (a) $\frac{7}{25} + \frac{24}{25}i$; (b) $\frac{21}{5} - \frac{12}{5}i$; (c) $3 + 4i$; (d) $-\frac{8}{29} + 0i$; (e) $\frac{22}{5}i$; (f) 15

Sol. (a) $\left(\frac{1+2i}{2+i} \times \frac{2-i}{2-i}\right)^2 = \frac{7}{25} + \frac{24}{25}i$

(b) $\frac{-i(9+6i)}{2-i} \Rightarrow \frac{-9i - 6i^2}{2-i} \Rightarrow \frac{6-9i}{2-i}$

Multiplying and dividing by $2 + i$

$$\frac{6-9i}{2-i}\times\frac{2+i}{2+i}$$

$$\Rightarrow\frac{(6-9i)(2+i)}{(2)^2-(i)^2}$$

$$\Rightarrow\frac{12+6i-18i-9i^2}{4+1}$$

$$\Rightarrow\frac{12+9-12i}{5}$$

Thus the a + bi form is $\frac{21}{5}-\frac{12}{5}i$

(c) $\frac{(4i^3-i)^2}{(2i+1)^2}$

$$\frac{i^2(4i^2-1)^2}{(2i+1)^2}$$

$$i^2\frac{(2i+1)^2(2i-1)^2}{(2i+1)^2} \qquad \therefore i^2=-1$$

$$=(-1)(2i-1)^2$$

$$=(-1)(4i^2+1-4i)$$

$$=(-1)(-4+1-4i)$$

$$=3+4i$$

(d) $\frac{3+2i}{2-5i}+\frac{3-2i}{2+5i}$

$$=\frac{(3+2i)(2+5i)+(2-5i)(3-2i)}{(2-5i)(2+5i)}$$

$$= \frac{6+15i+4i+10i^2+6-4i-15i}{4-25i^2}$$

$$= \frac{12+20i^2}{4-25i^2}$$

$$= \frac{12+20(-1)}{4-25(-1)} \qquad \ldots\ldots\left[\because i^2=-1\right]$$

$$= \frac{-8}{29}$$

$$\therefore \frac{3+2i}{2-5i}+\frac{3-2i}{2+5i}=\frac{-8}{29}+0i$$

$$\therefore a=\frac{-8}{29} \text{ and } b=0$$

(f) Given, four point $P_1P_2P_3P_4$ on a complex plane using which a square is drawn.

P1 ≡ (1, 0), $P_3 \equiv (3, 0)$

$|P_1P_3| = 2$ units = diagonal of the square

Now side of the square will be $\sqrt{2}$ units

P_2, P_4 can be interchangebly placed. It does not matter in this question.

Now coordinates of P_2 are $\left(1+\sqrt{2}\times\frac{1}{\sqrt{2}}, -\sqrt{2}\times\frac{1}{\sqrt{2}}\right)=(2,-1)$

Similarly that of P_4 are $\left(1+\sqrt{2}\times\frac{1}{\sqrt{2}}, +\sqrt{2}\times\frac{1}{\sqrt{2}}\right)=(2,1)$

Now, we have to find $(x_1+iy_1)(x_2+iy_2)(x_3+iy_3)(x_4+iy_4)$

$$=(1+0)(2-i)(3+0)(2+i)$$

$$=1\times3\times2(2-i)(2+i)$$

$$=3\times(4+1)$$

$$= 3 \times 5$$

$$= 15$$

16. Given that $x, y \in R$, solve :

दिया गया है : $x, y \in R$ सरल कीजिये :

(a) $(x + 2y) + i(2x - 3y) = 5 - 4i$ (b) $(x + iy) + (7 - 5i) = 9 + 4i$

(c) $x^2 - y^2 - i(2x + y) = 2i$ (d) $(2 + 3i)x^2 - (3 - 2i)y = 2x - 3y + 5i$

Ans. (a) $x = 1, y = 2$; (b) (2, 9); (c) $(-2, 2)$ or $\left(-\frac{2}{3}, -\frac{2}{3}\right)$; (d) $(1, 1)\left(0, \frac{5}{2}\right)$

Sol. (a) $(x + 2y) + i(2x - 3y) = 5 - 4i$

comparing real parts and imaginary parts we get $x + 2y = 5$ and $2x - 3y = -4$ _____(1)

$\Rightarrow x = 5 - 2y$

substituting value of x is (ii), we get $2(5 - 2y) - 3y = -4$

$10 - 4y - 3y = -4$

$\Rightarrow 7y = 14$

$\Rightarrow y = 2$

from (1), $x = 5 - 2(2) = 5 - 4$

$\therefore x = 1$

since $x = 1, y = 2$

(b) $x + iy + 7 - 5i = 9\ 4i$

$x + 7 = 9$ $y - 5 = 4$

$x = 2$ $y = 9$

(c) $x^2 - y^2 - i(2x + y) = 2i$

$x^2 - y^2 = 0$ $-(2x + y) = 2$

$x = y, x = -y$

$x = -2, y = 2$

$x = \frac{-2}{3}, \quad y = \frac{-2}{3}$

(d) $(2 + 3i)x^2 - (3 - 2i)y = 2x - 3y + 5i$

$\Rightarrow 2x^2 - 3y = 2x - 3y$

$\Rightarrow x^2 - x = 0$

$\Rightarrow x = 0, 1$ and $3x^2 + 2y = 5$

$\Rightarrow$ If $x = 0,\ y = \frac{5}{2}$ and $x = 1, y = 1$

are two solutions of the given equation which can also be represented as $\left(0, \frac{5}{2}\right)$ and $(1, 1)$, $\left(0, \frac{5}{2}\right)(1, 1)$

17. For what real values of x and y are the numbers $-3 + ix^2y$ and $x^2 + y + 4i$ conjugate complex?

x तथा y के किन वास्तविक मानों के लिये संख्यायें $-3 + ix^2y$ & $x^2 + y + 4i$ संयुग्म सम्मिश्र है?

Ans. $x = 1,\ y = -4$ or $x = -1,\ y = -4$

Sol. Given, $-3 + ix^2y = \overline{x^2 + y + 4i}$

$-3 + ix^2y = x^2 + y - 4i$

$-3 = x^2 + y$ and $x^2y = -4$

$\therefore -3 = x^2 - \frac{4}{x^2}$ [Putting $y = \frac{-4}{x^2}$ from (2) in (1)]

$x^4 + 3x^2 - 4 = 0$

$(x^2 + 4)(x^2 - 1) = 0$

$x^2 - 1 = 0$

$x = \pm 1$

From (2), $y = -4$, when $x = \pm 1$.

Hence, $x = 1, y = -4$ or $x = -1, y = 4$.

18. Given that $x, y \in R$, solve :

दिया गया है कि $x, y \in R$, हल कीजिये :

(a) $(x + 2y) + i(2x - 3y) = 5 - 4i$ (b) $\frac{x}{1+2i} + \frac{y}{3+2i} = \frac{5+6i}{8i-1}$

(c) $x^2 - y^2 - i(2x + y) = 2i$ (d) $(2 + 3i)x^2 - (3 - 2i)y = 2x - 3y + 5i$

Ans. (a) $x = 1, y = 2$; (b) $x = 1$ and $y = 2$; (c) $(-2, 2)$ or $\left(-\frac{2}{3}, -\frac{2}{3}\right)$; (d) $(1, 1)\left(0, \frac{5}{2}\right)$

Sol. $(x + 2y) + i(2x - 3y) = 5 - 4i$ comparing real parts and imaginary parts we get

$x + 2y = 5$ and $2x - 3y = -4$ _____(1)

$\Rightarrow x = 5 - 2y$

Substituting value of x is (ii), we get $2(5 - 2y) - 3y = -4$

$10 - 4y - 3y = -4$

$\Rightarrow 7y = 14$

$\Rightarrow y = 2$

from (1), $x = 5 - 2(2) = 5 - 4$

$\therefore x = 1$

Since $x = 1, y = 2$

(b) $\frac{x}{1+2i} + \frac{y}{3+2i} = \frac{5+6i}{-1+8i}$

$\therefore \frac{(3+2i)x + (1+2i)y}{(1+2i)(3+2i)} = \frac{5+6i}{-1+8i}$

$\therefore \frac{(3+2i)x + (1+2i)y}{3+2i+6i+4i^2} = \frac{5+6i}{-1+8i}$

$\therefore \frac{(3+2i)x + (1+2i)y}{3+8i-4} = \frac{5+6i}{-1+8i}$

$...\left[\because i^2 = -1\right]$

$\therefore \frac{3x + 2ix + y + 2iy}{-1+8i} = \frac{5+6i}{-1+8i}$

$\therefore (3x + y) + (2x + 2y)i = 5 + 6i$

Equating the real and imaginary parts separately, we get,

$3x + y = 5$ _____(1)

and $2x + 2y = 6$

i.e., $x + y = 3$ _____(2)

On subtracting, we get,

$2x = 2$

$\therefore x = 1$

$\therefore$ from (2), $1 + y = 3$

$\therefore\ y = 2$

Hence, $x = 1, y = 2$

(c) $x^2 - y^2 - i(2x + y) = 2i$

$x^2 - y^2 = 0 \qquad -(2x + y) = 2$

$x = y, x = -y$

$x = -2, y = 2$

$x = \frac{-2}{3},\ y = \frac{-2}{3}$

(d) $(2 + 3i)x^2 - (3 - 2i)y = 2x - 3y + 5i$

$\Rightarrow 2x^2 - 3y = 2x - 3y$

$\Rightarrow x^2 - x = 0$

$\Rightarrow x = 0, 1$ and $3x^2 + 2y = 5$

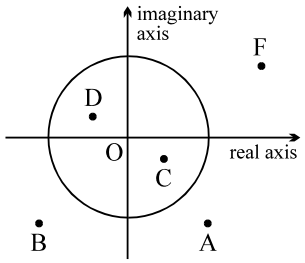
$\Rightarrow$ if $x = 0,\ y = \frac{5}{2}$ and if $x = 1, y = 1$

$\therefore\ x = 0,\ y = \frac{5}{2}$ and $x = 1, y = 1$ are two solutions of the given equation which can also be represented as

$\left(0, \frac{5}{2}\right)$ and $(1, 1)$, $\left(0, \frac{5}{2}\right)(1, 1)$

1. The digram shows several numbers in the complex plane. The circle is the unit circle centered at the origin. One of these numbers is the reciprocal of F, which is :

चित्र में सम्मिश्र समतल में कुछ संख्याऐं दर्शायी गयी है तथा वृत्त एक इकाई वृत्त है। जिसका केन्द्र मूल बिन्दु है। तब निम्न में से कौनसा बिन्दु F का प्रतिलोम होगा :



(A) A (B) B (C) C (D) D

Ans. C

Sol. Let F as $a + bi$, $a, b \in R$

where we see from the diagram that $a, b > 0$ and $a^2 + b^2 > 1$ (as F lies outside the unit circle)

Since $\alpha = \frac{1}{a+bi} = \frac{a-bi}{a^2+b^2} = \frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i$,

(real part + ve and imaginary part – ve and both less than unity)

we see that the reciprocal of F is in IV quadrant, since the real part is positive and the imaginary part is negative. Also, the magnitude of the reciprocal is

$$\frac{1}{a^2+b^2}\sqrt{a^2+(-b)^2} = \frac{1}{\sqrt{a^2+b^2}} < 1$$

Thus the only possibility is point C.

2. If $z = x + iy$ and $\omega = \frac{1-iz}{z-i}$ then $|\omega| = 1$ implies that, in the complex plane :

(A) z lies on the imaginary axis (B) z lies on the real axis

(C) z lies on the unit circle (D) None of these

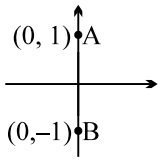
यदि $z = x + iy$ तथा $\omega = \frac{1 - iz}{z - i}$ है, तो $|\omega| = 1$ का तात्पर्य है, समिश्रतल में :

(A) z काल्पनिक अक्ष पर है (B) z वास्तविक अक्ष पर है

(C) z इकाई वृत्त पर है (D) इनमें से कोई नहीं

Ans. B

Sol. $w = \frac{-i(z+i)}{z-i}$; $|w| = \left|\frac{z+i}{z-i}\right| = 1 \quad \Rightarrow \quad |z+i| = |z-i|$



$\Rightarrow$ z lies on the perpendicular bisector of the segment joining (0, 1) and (0, – 1) which is x-axis

$\Rightarrow$ z lies on x-axis

$\Rightarrow$ Im (z) is real

3. On the complex plane locus of a point z satisfying the inequality

$2 \le |z - 1| < 3$ denotes

(A) region between the concentric circles of radii 3 and 1 centered at (1, 0)

(B) region between the concentric circles of radii 3 and 2 centered at (1, 0) excluding the inner and outer boundaries.

(C) region between the concentric circles of radii 3 and 2 centered at (1, 0) including the inner and outer boundaries.

(D) region between the concentric circles of radii 3 and 2 centered at (1, 0) including the inner boundary and excluding the outer boundary.

सम्मिश्र तल में z का बिन्दुपथ असमिका $2 \le |z - 1| < 3$ को सन्तुष्ट करता है :

(A) (1, 0) केन्द्र और त्रिज्याऐं 3 और 1 के सहकेन्द्रीय वृत्त के मध्य का क्षेत्र

(B) (1, 0) केन्द्र और त्रिज्याऐं 3 और 2 के सहकेन्द्रीय वृत्त के मध्य का क्षेत्र अन्तः और बाह्य परिसीमा को छोडकर

(C) (1, 0) केन्द्र और त्रिज्याऐं 3 और 2 के सहकेन्द्रीय वृत्त के मध्य का क्षेत्र अन्तः और बाह्य परिसीमा के साथ

(D) (1, 0) केन्द्र और त्रिज्याऐं 3 और 2 के सहकेन्द्रीय वृत्त के मध्य का क्षेत्रअन्तः परिसीमा के साथ और बाह्य परिसीमा को छोडकर

Ans. D

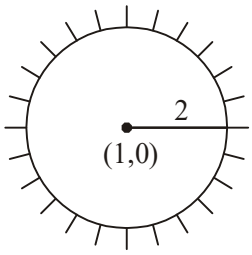
Sol. $|z - 1| \ge 2$

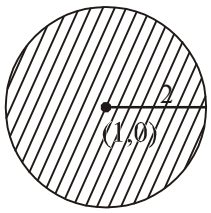
$z \rightarrow$ represnets boundary as well as outer region of circle

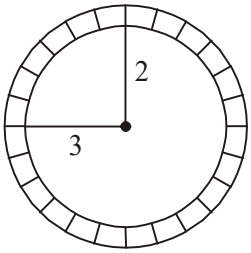
$|z - 1| < 3$

$z \rightarrow$ inner region of circle

c(1, 0), r = 3







4. Let $Z_1 = (8 + i)\sin\theta + (7 + 4i)\cos\theta$ and $Z_2 = (1 + 8i)\sin\theta + (4 + 7i)\cos\theta$ are two complex numbers. If $Z_1 \cdot Z_2 = a + ib$ where $a, b \in R$ then the largest value of $(a + b)\ \forall\ \theta \in R$, is :

माना $Z_1 = (8 + i)\sin\theta + (7 + 4i)\cos\theta$ और $Z_2 = (1 + 8i)\sin\theta + (4 + 7i)\cos\theta$ दो सम्मिश्र संख्याऐं है। यदि $Z_1 \cdot Z_2 = a + ib$ जहां $a, b \in R$ तब $(a + b)\ \forall\ \theta \in R$ का अधिकतम मान है :

(A) 75 (B) 100 (C) 125 (D) 130

Ans. C

Sol. $Z_1 = (8\sin\theta + 7\cos\theta) + i\,(\sin\theta + 4\cos\theta)$

$Z_2 = (\sin\theta + 4\cos\theta) + i\,(8\sin\theta + 7\cos\theta)$

hence $\left.\begin{matrix} Z_1 = x + iy \\ Z_2 = y + ix \end{matrix}\right]$ where $x = (8\sin\theta + 7\cos\theta)$ and $y = (\sin\theta + 4\cos\theta)$

$Z_1 \cdot Z_2 = (xy - xy) + i(x^2 + y^2) = 0 \quad \Rightarrow \quad a = 0;\ b = x^2 + y^2$

now, $x^2 + y^2 = (8\sin\theta + 7\cos\theta)^2 + (\sin\theta + 4\cos\theta)^2$

$= 65\sin^2\theta + 65\cos^2\theta + 120\sin\theta \cdot \cos\theta$

$= 65 + 60\sin 2\theta$

hence $Z_1 \cdot Z_2\big|_{max} = 125$

5. The locus of z, for $\arg z = -\ \pi/3$ is :

(A) same as the locus of z for $\arg z = 2\pi/3$

(B) same as the locus of z for $\arg z = \pi/3$

(C) the part of the straight line $\sqrt{3}\,x + y = 0$ with $(y < 0, x > 0)$

(D) the part of the straight line $\sqrt{3}\,x + y = 0$ with $(y > 0, x < 0)$

यदि $\arg z = -\pi/3$ है तब z का बिन्दु पथ है :

(A) वहि जबकि $\arg z = 2\pi/3$ के लिए z का बिन्दु पथ

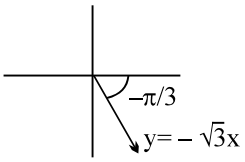
(B) वहि जबकि $\arg z = \pi/3$ के लिए z का बिन्दु पथ

(C) सरल रेखा $\sqrt{3}\,x + y = 0$ $(y<0, x>0)$ का भाग

(D) सरल रेखा $\sqrt{3}\,x + y = 0$ $(y > 0, x < 0)$ का भाग

Ans. C

Sol.



6. If P and Q are represented by the complex numbers z_1 and z_2 such that $\left|\frac{1}{z_1}+\frac{1}{z_2}\right| = \left|\frac{1}{z_1}-\frac{1}{z_2}\right|$, then the circumcentre of ΔOPQ (where O is the origin) is :

यदि P और Q सम्मिश्र संख्याऐं z_1 और z_2 को दर्शाते हैं इस प्रकार की $\left|\frac{1}{z_1}+\frac{1}{z_2}\right| = \left|\frac{1}{z_1}-\frac{1}{z_2}\right|$ तब ΔOPQ का परिकेन्द्र है :

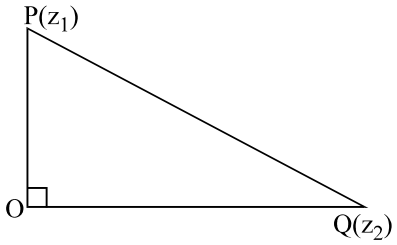
(जहां O मूल बिन्दु है)

(A) $\frac{z_1 - z_2}{2}$ (B) $\frac{z_1 + z_2}{2}$ (C) $\frac{z_1 + z_2}{3}$ (D) $z_1 + z_2$

Ans. B

Sol. We have $\left|\frac{1}{z_2}+\frac{1}{z_1}\right| = \left|\frac{1}{z_2}-\frac{1}{z_1}\right|$

$\Rightarrow |z_1 + z_2| = |z_1 - z_2|$ $\Rightarrow z_1\bar{z}_2 + z_2\bar{z}_1 = 0$



$\Rightarrow \dfrac{z_1}{z_2}$ is purely imaginary.

Hence ΔPQR is right angled at O.

$\therefore$ Circumcentre of ΔPOQ is the mid point of PQ i.e. $\dfrac{1}{2}(z_1 + z_2)$

7. The value of $e\left(CiS(-i) - CiS(i)\right)$ is equal to :

$e\left(CiS(-i) - CiS(i)\right)$ का मान बराबर होगा :

(A) 0 (B) $1 - e$ (C) $e - \dfrac{1}{e}$ (D) $e^2 - 1$

Ans. D

Sol. using $CiS\ \theta = e^{i\theta}$

$E = e\left[e^{-i^2} - e^{i^2}\right] = e[e - e^{-1}] = e^2 - 1$

8. All real numbers x which satisfy the inequality $\left|1 + 4i - 2^{-x}\right| \le 5$ where $i = \sqrt{-1}$, $x \in R$ are :

x के सभी वास्तविक मान जो असमिका $\left|1 + 4i - 2^{-x}\right| \le 5$ को सन्तुष्ट करते हैं। जहां $i = \sqrt{-1}$, $x \in R$:

(A) $[-2, \infty)$ (B) $(-\infty, 2]$ (C) $[0, \infty)$ (D) $[-2, 0]$

Ans. A

Sol. $(1 - 2^{-x})^2 + 16 \le 25$; $(1 - 2^{-x}) - 3^2 \le 0$; $(4 - 2^{-x})(-2 - 2^{-x}) \le 0$
$(2^{-x} - 4)(2^{-x} + 2) \ge 0$

9. Number of real or purely imaginary solution of the equation, $z^3 + iz - 1 = 0$ is :

(A) zero (B) one (C) two (D) three

समीकरण $z^3 + iz - 1 = 0$ के वास्तविक या पूर्णतः काल्पनिक हलों की कुल संख्या हैं :

(A) शून्य (B) एक (C) दो (D) तीन

Ans. A

Sol. Let x be the real solution .

$\Rightarrow x^3 - 1 + xi = 0 \Rightarrow x^3 - 1 = 0$ & $x = 0$ which is not possible

note that the equation has no purely imaginary root as well.

10. A point 'z' moves on the curve $|z-4-3i|=2$ in an argand plane. The maximum and minimum values of $|z|$ are :

यदि 'z', वक्र $|z-4-3i|=2$ पर स्थित है, तब $|z|$ का अधिकतम व न्यूनतम मान है :

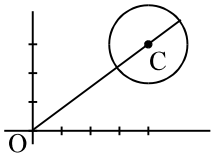
(A) 2, 1 (B) 6, 5 (C) 4, 3 (D) 7, 3

Ans. D

Sol. $|(x-4)+i(y-3)|=2$

circle with centre (4, 3) and radius 2 ;

Hence OC = 5



$|z|_{max}=5+2=7$

$|z|_{min}=5-2=3$

11. If z is a complex number satisfying the equation $|z+i|+|z-i|=8$, on the complex plane then maximum value of $|z|$ is :

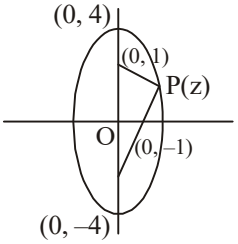
यदि z सम्मिश्र तल में समीकरण $|z+i|+|z-i|=8$ को सन्तुष्ट करता है तब $|z|$ का अधिकतम मान होगा :

(A) 2 (B) 4 (C) 6 (D) 8

Ans. B

Sol. If $|z+i|+|z-i|=8$,

$PF_1+PF_2=8$



$\therefore \quad |z|_{max}=4 \quad \Rightarrow \quad$ (B)

12. If $\left(\sqrt{3}+i\right)^{100}=2^{99}(a+ib)$, then a^2+b^2 is equal to :

यदि $\left(\sqrt{3}+i\right)^{100}=2^{99}(a+ib)$, तो a^2+b^2 का मान होगा :

(A) 2 (B) 1 (C) 3 (D) 4

Ans. D

Sol. $\left(2e^{\frac{i\pi}{6}}\right)^{100} = 2^{99}(p+iq)$

$$2^{100}\left(\cos\frac{50\pi}{3}+i\sin\frac{50\pi}{3}\right)=2^{99}(p+iq)$$

$$p+iq=2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)$$

$$P=-1,\ q=\sqrt{3}$$

13. If amp $(z_i)=\theta_i$, i = 1, 2, 3; then amp $\left(\frac{z_1}{z_2\,\bar{z}_3}\right)$ is equal to :

यदि कोणांक $(z_i)=\theta_i$, i = 1, 2, 3; तो, कोणांक $\left(\frac{z_1}{z_2\,\bar{z}_3}\right)$ बराबर है :

(A) $\frac{\theta_1}{\theta_2\theta_3}$ (B) $\frac{\theta_1\theta_2}{\theta_3}$ (C) $\theta_1-\theta_2-\theta_3$ (D) $\theta_1-\theta_2+\theta_3$

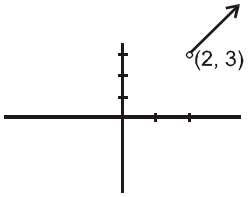
Ans. D

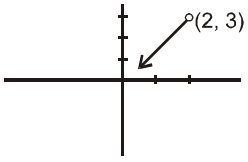
Sol. $\text{amp}\left(\bar{2}_3\right)=-\theta_3$

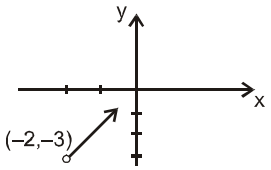
$$\text{amp}\left(\frac{z_1}{z_2\bar{z}_3}\right)=\theta_1-\theta_2+\theta_3$$

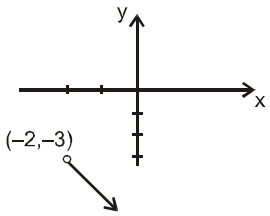
14. If $\text{Arg}(z-2-3i)=\frac{\pi}{4}$, then the locus of z is :

यदि $z-2-3i$ का कोणांक $\frac{\pi}{4}$ हो, तो z का बिन्दुपथ है :

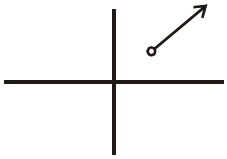
(A) 

(B) 

(C) 

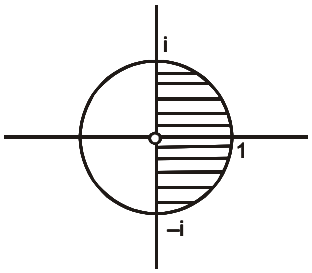
(D) 

Ans. A

Sol. 

15. The locus of z which lies in shaded region is best represented by :

z का बिन्दु पथ जो छायांकित क्षेत्र में स्थित हो, का श्रेष्ठ निरूपण है :



(A) $|z| \le 1, \frac{-\pi}{2} \le \arg z \le \frac{\pi}{2}$

(B) $|z| = 1, \frac{-\pi}{2} \le \arg z \le 0$

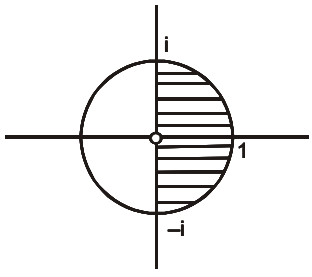
(C) $|z| \ge 0, \ 0 \le \arg z \le \frac{\pi}{2}$

(D) $|z| \le 1, \frac{\pi}{2} \le \arg z \le \pi$

Ans. A

Sol. Clearly

$$|z| \ge 1, \frac{-\pi}{2} \le \arg(z) \le \frac{\pi}{2}$$



16. Find the square root of :

निम्न का वर्ग मूल ज्ञात कीजिये :

(a) $9 + 40\,i$ (b) $-11 - 60\,i$

(c) $50\,i$

Ans. (a) $\pm(5 + 4i)$ (b) $\pm(5 - 6i)$ (c) $\pm 5(1 + i)$

Sol. (a) $(x + iy)^2 = 9 + 40i$

$\therefore\ x^2 - y^2 = 9$ _____(1)

and $xy = 20$ _____(2)

squing (1) and adding with 4 times the square of (2)

we get $x^4 + y^4 - 2x^2 y^2 + 4x^2 y^2 = 81 + 1600$

$\Rightarrow (x^2 + y^2)^2 = 168$

$\Rightarrow x^2 + y^2 = 4$ _____(3)

from (1) + (3) we get $x^2 = 25$

$\Rightarrow x = \pm 5$ and $y = 16$

$\Rightarrow y = \pm 4$

from equation (2) we can see that x and y are of same sign

$\therefore\ x + iy = +(5 + 4i)$ or $= (5 + 4i)$

$\therefore$ sq. roots of $a + 40i = \pm(5 + 4i)$

(b) Let, $(a + ib)^2 = -11 - 60i$

Now using, $(a + b)^2 = a^2 + b^2 + 2ab$

$\Rightarrow a^2 + (bi)^2 + 2abi = -11 - 60i$

Since $i^2 = -1$

$\Rightarrow a^2 - b^2 + 2abi = -11 - 60i$

Now, separating real and complex parts, we get

$\Rightarrow a^2 - b^2 = -11$ _____eq. (1)

$\Rightarrow 2ab = -60$ _____eq. (2)

$\Rightarrow\ a = -\dfrac{30}{b}$

Now, using the value of a in eq.(1), we get

$$\Rightarrow \left(-\frac{30}{b}\right)^2 - b^2 = -11$$

$\Rightarrow 900 - b^4 = -11b^2$

$\Rightarrow b^4 - 11b^2 - 900 = 0$

Simplify and get the value of b^2, we get,

$\Rightarrow b^2 = 36$ or $b^2 = -25$

as b is real no. so, $b^2 = 36$

$b = 6$ or $b = -6$

Therefore, $a = -5$ or $a = 5$

Hence the square root of the complex no. is $-5 + 6i$ and $5 - 6i$

(c) $\sqrt{50i} = a + ib$

$50i = a^2 - b^2 + 2abi$

$a^2 - b^2 = 0$ $\qquad$ $2ab = 50$

$a = b \mid a = -b$ $\qquad$ $ab = 25$

$$\sqrt{50i} = \pm 5(1+i)$$

17. (a) If $f(x) = x^4 + 9x^3 + 35x^2 - x + 4$, find $f(-5 + 4i)$
यदि $f(x) = x^4 + 9x^3 + 35x^2 - x + 4$, तो $f(-5 + 4i)$ का मान ज्ञात कीजिये।
(b) If $g(x) = x^4 - x^3 + x^2 + 3x - 5$, find $g(2 + 3i)$
यदि $g(x) = x^4 - x^3 + x^2 + 3x - 5$, तो $g(2 + 3i)$ का मान ज्ञात कीजिये।

Ans. (a) -160 (b) $-(77 + 108i)$

Sol. (a) $f(x) = x^4 + 9x^3 + 35x^2 - x + 4$

if $x = -5 + 4i$

$\Rightarrow (x + 5) = 4i$

$\Rightarrow (x + 5)^2 = (4i)^2$

$\Rightarrow X^2 + 25 + 10x = -16$

$\Rightarrow X^2 + 10x + 41 = 0$ $\qquad$ _____(1)

All we have to do is divide the expression $x^4 + 9x^3 + 35x^2 - x + 4$ by $x^2 + 10x + 41$ via the long division method.

$$\therefore x^4 + 9x^3 + 35x^2 - x + 4 = \left(x^2 + 10x + 41\right)$$

$$\therefore f(x) = x^4 + 9x^3 + 35x^2 - x + 4 = -160$$

(b) $x^4 - x^3 + x^2 + 3x - 5$

$(2+3i)^4 - (2+3i)^3 + (2+3i)^2 + 3(2+3i) - 5$

$(2+3i)^4 : -119 - 120i$

$(2+3i)^3 : -46 + 9i$

$(2+3i)^2 : -5 + 12i$

$= (-119 - 120i) + 46 - 9i + (-5 + 12i) + 6 + 9i$

$= -119 - 120i + 46 - 9i - 5 + 12i + 6 + 9i - 5$

$= -120i - 9i + 12i + 9i - 119 + 46 - 5 + 6 - 5$

$= -108i - 119 + 46 - 5 + 6 - 5$

$= -77 - 108i$

18. Simplify and express the result in the form of $a + bi$:
सरल कीजिये एवं अपने परिणाम को $a + bi$ के रूप में लिखियें :

(a) $\left(\frac{4i^3 - i}{2i + 1}\right)^2$ (b) $\frac{3+2i}{2-5i} + \frac{3-2i}{2+5i}$ (c) $\frac{(2+i)^2}{2-i} - \frac{(2-i)^2}{2+i}$

Ans. (a) $3 + 4i$ (b) $-\frac{8}{29} + 0i$ (c) $\frac{22}{5}i$

Sol. (a) $\left(\frac{1+2i}{2+i} \times \frac{2-i}{2-i}\right)^2 = \frac{7}{25} + \frac{24}{25}i$

(b) $\frac{-i(9+6i)}{2-i} \Rightarrow \frac{-9i - 6i^2}{2-i} \Rightarrow \frac{6-9i}{2-i}$

Multiplying and dividing by $2 + i$

$\frac{6-9i}{2-i} \times \frac{2+i}{2+i}$

$\Rightarrow \frac{(6-9i)(2+i)}{(2)^2 - (i)^2}$

$\Rightarrow \frac{12 + 6i - 18i - 9i^2}{4+1}$

$\Rightarrow \frac{12+9-12i}{5}$

Thus the a + bi form is $\frac{21}{5}-\frac{12}{5}i$

(c) $\frac{\left(4i^3-i\right)^2}{\left(2i+1\right)^2}$

$\frac{i^2\left(4i^2-1\right)^2}{\left(2i+1\right)^2}$

$i^2\frac{\left(2i+1\right)^2\left(2i-1\right)^2}{\left(2i+1\right)^2}$ $\quad\quad\quad

$$\therefore \frac{3+2i}{2-5i}+\frac{3-2i}{2+5i}=\frac{-8}{29}+0i$$

$$\therefore a=\frac{-8}{29} \text{ and } b=0$$

(f) Given, four point $P_1P_2P_3P_4$ on a complex plane using which a square is drawn.

P1 ≡ (1, 0), $P_3 \equiv (3, 0)$

$|P_1P_3| = 2$ units = diagonal of the square

Now side of the square will be $\sqrt{2}$ units

P_2, P_4 can be interchangebly placed. It does not matter in this question.

Now coordinates of P_2 are $\left(1+\sqrt{2}\times\frac{1}{\sqrt{2}},-\sqrt{2}\times\frac{1}{\sqrt{2}}\right)=(2,-1)$

Similarly that of P_4 are $\left(1+\sqrt{2}\times\frac{1}{\sqrt{2}},+\sqrt{2}\times\frac{1}{\sqrt{2}}\right)=(2,1)$

Now, we have to find $(x_1+iy_1)(x_2+iy_2)(x_3+iy_3)(x_4+iy_4)$

$=(1+0)(2-i)(3+0)(2+i)$

$=1\times3\times2(2-i)(2+i)$

$=3\times(4+1)$

$= 3 \times 5$

$= 15$

19. Find the modulus, argument and the principal argument of the complex number $z=1+\cos\left(\frac{10\pi}{9}\right)+i\sin\left(\frac{10\pi}{9}\right)$:

सम्मिश्र संख्या $z = 1 + \cos\left(\frac{10\pi}{9}\right) + i\sin\left(\frac{10\pi}{9}\right)$ के लिये मापांक, कोणांक तथा मुख्य कोणांक ज्ञात कीजिए :

Ans. Principal Arg $z=-\frac{4\pi}{9}$; $|z| = 2\cos\frac{4\pi}{9}$; Arg $z = 2k\pi-\frac{4\pi}{9}$ $k \in I$

Sol. $z=1+\frac{10\pi}{9}+i\sin\frac{10\pi}{9}$

$$z = 2\cos^2\frac{5\pi}{9} + 2i\sin\frac{5\pi}{9}\cos\frac{5\pi}{9}$$

$$\Rightarrow z = -2\cos\frac{4\pi}{9}\left[\cos\frac{5\pi}{9} + i\sin\frac{5\pi}{9}\right]$$

$$= 2\cos\frac{4\pi}{9}\left[-\cos\frac{5\pi}{9} - i\sin\frac{5\pi}{9}\right]$$

$$= 2\cos\frac{4\pi}{9}\left[\cos\left(\pi + \frac{5\pi}{9}\right) + i\sin\left(\pi + \frac{5\pi}{9}\right)\right]$$

$$\Rightarrow |z| = 2\cos\frac{4\pi}{9}$$

20. Find the modulus, argument and the principal argument of the complex number $(\tan 1 - i)^2$:

सम्मिश्र संख्या $(\tan 1 - i)^2$ के लिये मापांक, कोणांक तथा मुख्य कोणांक ज्ञात कीजिए :

Ans. Modulus = $\sec^2 1$, Arg = $2n\pi + (2 - \pi)$, Principal Arg = $(2 - \pi)$

Sol. Modulus = $\sec^2 1$, arg = $2k\pi + (2 - \pi)$, Principal Arg = $(2 - \pi)$

मापांक = $\sec^2 1$, कोणांक = $2k\pi + (2 - \pi)$, मुख्य मान कोणांक = $(2 - \pi)$

1. If z_1 and z_2 are two non-zero complex numbers such that $|z_1 + z_2| = |z_1| + |z_2|$, then $\text{Arg}\, z_1 - \text{Arg}\, z_2$ is equal to :

यदि z_1 और z_2 अशून्य सम्मिश्र संख्याऐं इस प्रकार है कि $|z_1 + z_2| = |z_1| + |z_2|$ तब $\text{Arg}\, z_1 - \text{Arg}\, z_2$ बराबर :

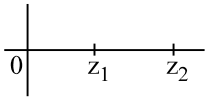
(A) $-\pi$ (B) $-\pi/2$ (C) 0 (D) $\pi/2$

Ans. C

Sol. $|z_1 + z_2| = |z| + |z_2|$

$\Rightarrow \quad \sqrt{(r_1\cos\theta_1 + r_2\cos\theta_2) + i(r_1\sin\theta_1 + r_2\sin\theta_2)} = r_1 + r_2$

$\sqrt{r_1^2 + r_2^2 + 2r_1r_2\cos(\theta_1 - \theta_2)} = r_1 + r_2$



This is possible only if $\theta_1 = \theta_2$

$\Rightarrow$ 0, z_1 and z_2 are collinear with z_1 and z_2 on the same side of the origin

$\Rightarrow$ $\text{Arg}\, z_1 = \text{Arg}\, z_2$

2. Let Z be a complex number satisfying the equation $(Z^3 + 3)^2 = -16$ then $|z|$ has the value equal to :

यदि सम्मिश्र संख्या Z समीकरण $(Z^3 + 3)^2 = -16$ को सन्तुष्ट करता है तब $|z|$ बराबर :

(A) $5^{1/2}$ (B) $5^{1/3}$ (C) $5^{2/3}$ (D) 5

Ans. B

Sol. $(Z^3 + 3)^2 = 16i^2$

$Z^3 + 3 = 4i$ or $-4i$

$Z^3 = -3 + 4i$ or $-3 - 4i$

$|Z|^3 = |-3 + 4i| = 5$

$|Z|^3 = 5 \quad \Rightarrow \quad |Z| = 5^{1/3}$

3. If z_1, z_2, z_3 are 3 distinct complex numbers such that $\frac{3}{|z_2 - z_3|} = \frac{4}{|z_3 - z_1|} = \frac{5}{|z_1 - z_2|}$, then the value of $\frac{9}{z_2 - z_3} + \frac{16}{z_3 - z_1} + \frac{25}{z_1 - z_2}$ equals :

यदि z_1, z_2, z_3 तीन भिन्न सम्मिश्र संख्याऐं इस प्रकार है कि $\frac{3}{|z_2 - z_3|} = \frac{4}{|z_3 - z_1|} = \frac{5}{|z_1 - z_2|}$ तब $\frac{9}{z_2 - z_3} + \frac{16}{z_3 - z_1} + \frac{25}{z_1 - z_2}$ बराबर :

(A) 0 (B) 3 (C) 4 (D) 5

Ans. A

Sol. We have $\frac{3}{|z_2 - z_3|} = \frac{4}{|z_3 - z_1|} = \frac{5}{|z_1 - z_2|} = k$ (let)

$\Rightarrow \quad \frac{9}{|z_2 - z_3|^2} = \frac{16}{|z_3 - z_1|^2} = \frac{25}{|z_1 - z_2|^2} = k^2$

Now $\frac{9}{|z_2 - z_3|^2} = k^2 \quad \Rightarrow \quad \frac{9}{z_2 - z_3} = k^2(\bar{z}_2 - \bar{z}_3)$ ....(1) [As $|z|^2 = z\bar{z}$]

|||ly $\frac{16}{|z_3 - z_1|^2} = k^2 \quad \Rightarrow \quad \frac{16}{z_3 - z_1} = k^2(\bar{z}_3 - \bar{z}_1)$ ....(2)

|||ly $\frac{25}{|z_1 - z_2|^2} = k^2 \quad \Rightarrow \quad \frac{25}{z_1 - z_2} = k^2(\bar{z}_1 - \bar{z}_2)$ ....(3)

$\therefore$ On adding (1), (2) and (3), we get

$$\frac{9}{z_2 - z_3} + \frac{16}{z_3 - z_1} + \frac{25}{z_1 - z_2} = k^2(\bar{z}_2 - \bar{z}_3 + \bar{z}_3 - \bar{z}_1 + \bar{z}_1 - \bar{z}_2) = 0$$

4. The area of the triangle whose vertices are the roots of $z^3 + iz^2 + 2i = 0$ is :

उस त्रिभुज का क्षेत्रफल जिसके शीर्ष समीकरण $z^3 + iz^2 + 2i = 0$ के मूल है, होगा :

(A) 2 (B) $\frac{3}{2}\sqrt{7}$ (C) $\frac{3}{4}\sqrt{7}$ (D) $\sqrt{7}$

Ans. A

Sol. $(z - i)(z^2 + 2iz - 2) = 0 \Rightarrow \quad z = -i, (1 - i), -(1 + i)$

$$\text{Area} = \frac{1}{2}\begin{vmatrix} 0 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & -1 & 1 \end{vmatrix} = |-2| = 2$$

5. Consider two complex numbers α and β as $\alpha = \left(\frac{a+bi}{a-bi}\right)^2 + \left(\frac{a-bi}{a+bi}\right)^2$, where a, b $\in$ R and $\beta = \frac{z-1}{z+1}$, where $|z| = 1$, then :

(A) Both α and β are purely real
(B) Both α and β are purely imaginary
(C) α is purely real and β is purely imaginary
(D) β is purely real and α is purely imaginary

माना α और β दो सम्मिश्र संख्याऐं है। इस प्रकार कि $\alpha = \left(\frac{a+bi}{a-bi}\right)^2 + \left(\frac{a-bi}{a+bi}\right)^2$, जहां a, b $\in$ R और $\beta = \frac{z-1}{z+1}$, जहां $|z| = 1$, तब :

(A) दोनों α और β पूर्णतः वास्तविक है
(B) दोनों α और β पूर्णतः काल्पनिक है
(C) α पूर्णतः वास्तविक व β पूर्णतः काल्पनिक है
(D) β पूर्णतः वास्तविक तथा α पूर्णतः काल्पनिक है

Ans. C

Sol. Note that $\alpha = \overline{\alpha} \Rightarrow$ α is real

and $\beta + \overline{\beta} = \frac{z-1}{z+1} + \frac{\overline{z}-1}{\overline{z}+1} = \frac{(z-1)(\overline{z}+1)+(z+1)(\overline{z}-1)}{(z+1)(\overline{z}+1)} = \frac{2z\overline{z}-2}{D^r} = 0$

as $z\overline{z} = |z|^2 = 1$ (given)

6. Let Z is complex satisfying the equation $z^2 - (3+i)z + m + 2i = 0$, where $m \in R$. Suppose the equation has a real root. The additive inverse of non real root, is :

माना Z कोई सम्मिश्र संख्या है जो समीकरण $z^2 - (3 + i)z + m + 2i = 0$ को सन्तुष्ट करता है, जहां $m \in R$. माना समीकरण का एक मूल वास्तविक है तब अवास्तविक मूल का योग प्रतिलोम होगा :

(A) $1 - i$ (B) $1 + i$ (C) $-1 - i$ (D) -2

Ans. C

Sol. Let α be the real root

$\alpha^2 - (3 + i)\alpha + m + 2i = 0$

$(\alpha^2 - 3\alpha + m) + i(2 - \alpha) = 0$

$\therefore \alpha = 2$ (real root)

$\therefore \quad 4 - 6 + m = 0 \Rightarrow m = 2$

Product of the roots = $2(1 + i)$ with one root as 2

non real root = $1 + i$, addivitve inverse is $-1 - i$

7. The minimum value of $|z-1+2i|+|4i-3-z|$ is :

$|z-1+2i|+|4i-3-z|$ का न्यूनतम मान होगा :

(A) $\sqrt{5}$ (B) 5 (C) $2\sqrt{13}$ (D) $\sqrt{15}$

Ans. C

Sol. The expression is the sum of the distance of z from the two points $1-2i$ and $-3+4i$. The minimum value is the distance between these two points = $\sqrt{4^2+6^2} = 2\sqrt{13}$

8. If $z=\frac{\pi}{4}(1+i)^4\left(\frac{1-\sqrt{\pi}\,i}{\sqrt{\pi}+i}+\frac{\sqrt{\pi}-i}{1+\sqrt{\pi}\,i}\right)$ then $\left(\frac{|z|}{\text{amp}\,z}\right)$ equals :

यदि $z=\frac{\pi}{4}(1+i)^4\left(\frac{1-\sqrt{\pi}\,i}{\sqrt{\pi}+i}+\frac{\sqrt{\pi}-i}{1+\sqrt{\pi}\,i}\right)$ हो, तो $\left(\frac{|z|}{\text{amp}\,z}\right)$ बराबर होगा :

(A) 1 (B) π (C) 3π (D) 4

Ans. D

Sol. $z=\frac{\pi}{4}(1+i)^4\left(\frac{(1+\pi)+(\pi+1)}{\sqrt{\pi}+\pi i+i-\sqrt{\pi}}\right)$; $z=\frac{\pi}{2}(1+i)^4\cdot\frac{1}{i}$

$\therefore \quad |z|=\frac{\pi}{2}\cdot 4=2\pi; \quad \text{amp.}(z)=0+4\cdot\frac{\pi}{4}-\frac{\pi}{2} \quad \Rightarrow \quad \frac{|z_1|}{\text{amp.}(z)}=4$

9. A particle starts from a point $z_0=1+i$, where $i=\sqrt{-1}$. It moves horizontally away from origin by 2 units and then vertically away from origin by 3 units to reach a point z_1. From z_1 particle moves $\sqrt{5}$ units in the direction of $2\hat{i}+\hat{j}$ and then it moves through an angle of $\text{cosec}^{-1}\sqrt{2}$ in anticlockwise direction of a circle with centre at origin to reach a point z_2. The arg z_2 is given by :

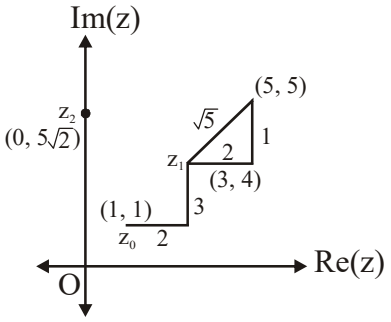
एक कण बिन्दु $z_0 = 1 + i$ से गति करना प्रारम्भ करता है जहाँ $i = \sqrt{-1}$ यह मूल बिन्दु से क्षैतिज दिशा में 2 मात्रक दूर व उर्द्धाधर दिशा में 3 मात्रक दूर बिन्दु z_1 तक पहुंचता है, बिन्दु z_1 से कण $2\hat{i}+\hat{j}$ दिशा में $\sqrt{5}$ मात्रक गति करता है तत्पश्चात् यह वामावर्त्त दिशा में एक वृत्ताकार पथ पर, यदि वृत्त का केन्द्र मूल बिन्दु है, कोण $\text{cosec}^{-1}\sqrt{2}$ तय करता है बिन्दु z_2 तक पहुंचता है। तो arg z_2 होगा :

(A) $\sec^{-1}2$ (B) $\cot^{-1}0$ (C) $\sin^{-1}\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)$ (D) $\cos^{-1}\left(\frac{-1}{2}\right)$

Ans. B

Sol. Clearly $z_1 = 3 + 4i$

After moving by $\sqrt{5}$ distance in direction of $2\hat{i} + \hat{j}$, particle will reach at point $\left(5\hat{i} + 5\hat{j}\right)$



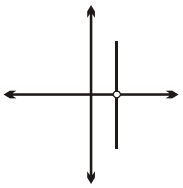
If particle moves by an angle $\frac{\pi}{4}$ then it will reach at y-axis

At $z_2 = 0 + 5\sqrt{2}\,i$

Hence $\text{amp}(z_2) = \frac{\pi}{2} = \cot^{-1}0$

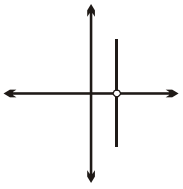
10. Let $z = x + iy$ then locus of moving point P(z) such that $\frac{1+\bar{z}}{z} \in R$, is : (where $i^2 = -1$)

माना $z = x + iy$ तब चर बिन्दु P(z) का बिन्दु पथ इस प्रकार कि $\frac{1+\bar{z}}{z} \in R$ है (जहां $i^2 = -1$) :



(A) union of lines with equations $x = 0$ and $y = -1/2$ but excluding origin
(B) union of lines with equations $x = 0$ and $y = 1/2$ but excluding origin
(C) union of lines with equations $x = -1/2$ and $y = 0$ but excluding origin
(D) union of lines with equations $x = 1/2$ and $y = 0$ but excluding origin

माना $z = x + iy$ तब चर बिन्दु P(z) का बिन्दु पथ इस प्रकार कि $\frac{1+\bar{z}}{z} \in R$ है (जहां $i^2 = -1$) :



(A) सरल रेखाओं $x = 0$ और $y = -1/2$ का संघ मूल बिन्दु को छोड़कर
(B) सरल रेखाओं $x = 0$ और $y = 1/2$ का संघ मूल बिन्दु को छोड़कर
(C) $x = -1/2$ और $y = 0$ रेखाओं का संघ मूल बिन्दु छोड़कर
(D) $x = 1/2$ और $y = 0$ रेखाओं का संघ मूल बिन्दु छोड़कर

Ans. C

Sol. Given $\frac{1+\bar{z}}{z}$ is real $\Rightarrow \frac{1+\bar{z}}{z} = \frac{1+z}{\bar{z}} \Rightarrow \bar{z}+\bar{z}^2 = z+z^2 \Rightarrow (\bar{z}-z)+(\bar{z}-z)(\bar{z}+z)=0$

$\Rightarrow$ $(\bar{z}-z)(1+\bar{z}+z)=0$, so either $\bar{z}=z\ (z\neq 0)$ or $z+\bar{z}+1=0$

$\Rightarrow$ $y=0$ or $x=\frac{-1}{2}$ but excluding origin.

11. Let C_1 and C_2 are concentric circles of radius 1 and 8/3 respectively having centre at $(3, 0)$ on the argand plane. If the complex number z satisfies the inequality, $\log_{1/3}\left(\frac{|z-3|^2+2}{11|z-3|-2}\right) > 1$ then :

(A) z lies outside C_1 but inside C_2 (B) z lies inside of both C_1 and C_2

(C) z lies outside both of C_1 and C_2 (D) None of these

माना C_1 और C_2 सम्मिश्र समतल में दो सहकेन्द्रिय वृत्त है जिनका केन्द्र (3, 0) और त्रिज्याऐं क्रमशः 1 और 8/3 है। यदि सम्मिश्र संख्या z असमिका $\log_{1/3}\left(\frac{|z-3|^2+2}{11|z-3|-2}\right) > 1$ को सन्तुष्ट करता है तब :

(A) z, C_1 के बाहर तथा C_2 के अन्दर है (B) z दोनों C_1 और C_2 के अन्दर है

(C) z, C_1 और C_2 दोनों के बाहर है (D) उपरोक्त में से कोई नहीं

Ans. A

Sol. note that $11|z-3|-2>0$

$\frac{|z-3|^2+2}{11|z-3|-2} < \frac{1}{3}$; put $|z-3|=t \Rightarrow (3t-8)(t-1)<0 \Rightarrow 1<|z-3|<8/3$

$\Rightarrow$ z lies between the two concentric circles

12. The equation $z\bar{z}+(4-3i)z+(4+3i)\bar{z}+5=0$ represents a circle whose radius is :

समीकरण $z\bar{z}+(4-3i)z+(4+3i)\bar{z}+5=0$ एक वृत्त को प्रदर्शित करती है तो वृत्त की त्रिज्या होगी :

(A) 5 (B) 5/2 (C) $2\sqrt{5}$ (D) $\sqrt{5}$

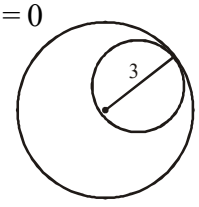
Ans. C

Sol. The $z = x + iy$

$\Rightarrow$ $x^2+y^2+(4-3i)(x+iy)+(4+3i)(x-iy)+5=0$

$\Rightarrow$ $x^2+y^2+8x+6y+5=0$, a circle

radius = $\sqrt{4^2+3^2-5} = 2\sqrt{5}$



13. If $z_1=-3+5i$; $z_2=-5-3i$ and z is a complex number lying on the line segment joining z_1 and z_2 then arg z can be :

$z_1 = -3 + 5i$; $z_2 = -5 - 3i$ को मिलाने वाली रेखा खण्ड पर स्थित बिन्दु z का arg z होगा :

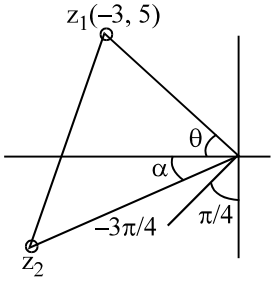
(A) $-\frac{3\pi}{4}$ (B) $-\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{5\pi}{6}$

Ans. D

Sol. $\tan\theta = \frac{5}{3} \Rightarrow \theta > \frac{\pi}{4}$

$\tan\alpha = \frac{3}{5} \Rightarrow \alpha < \frac{\pi}{4}$

$\Rightarrow$ A/B/C cannot be the answer



14. $z_1 = \frac{a}{1-i}$; $z_2 = \frac{b}{2+i}$; $z_3 = a - bi$ for a, b $\in$ R

if $z_1 - z_2 = 1$ then the centroid of the triangle formed by the points z_1, z_2, z_3 in the argand's plane is given by :

यदि $z_1 = \frac{a}{1-i}$; $z_2 = \frac{b}{2+i}$; $z_3 = a - bi$ जहाँ a, b $\in$ R

यदि $z_1 - z_2 = 1$ तब z_1, z_2 और z_3 द्वारा सम्मिश्र तल में निर्मित त्रिभुज का केन्द्रक है :

(A) $\frac{1}{9}(1 + 7i)$ (B) $\frac{1}{3}(1 + 7i)$ (C) $\frac{1}{3}(1 - 3i)$ (D) $\frac{1}{9}(1 - 3i)$

Ans. A

Sol. $z_1 = \frac{a(1+i)}{2}$; $z_2 = \frac{b(2-i)}{5}$

$\frac{a(1+i)}{2} - \frac{b(2-i)}{5} = 1$

$5a(1 + i) - 2b(2 - i) = 10$

$(5a - 4b - 10) + i(5a + 2b) = 0$

$5a - 4b = 10$; $5a = -2b$

$-6b = 10 \Rightarrow b = \frac{-5}{3}$

$5a = -2\left(\frac{-5}{3}\right) = \frac{10}{3} \Rightarrow a = \frac{2}{3}$

15. Number of complex numbers z such that $|z| = 1$ and $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$ is :

(A) 4 (B) 6 (C) 8 (D) More than 8

सम्मिश्र सख्याओं z की संख्या जिसके लिए $|z| = 1$ और $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$ है :

(A) 4 (B) 6 (C) 8 (D) 8 से अधिक

Ans. C

Sol. Let $z = \cos x + i \sin x,\ x \in [0, 2\pi)$. Then

$$1 = \left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = \frac{|z^2 + \bar{z}^2|}{|z|^2} = |\cos 2x + i \sin 2x + \cos 2x - i \sin 2x| = 2|\cos 2x|$$

hence $\cos 2x = 1/2$ or $\cos 2x = -1/2$

If $\cos 2x = 1/2$, then

$$x_1 = \frac{\pi}{6},\ x_2 = \frac{5\pi}{6},\ x_3 = \frac{7\pi}{6},\ x_4 = \frac{11\pi}{6}$$

If $\cos 2x = -\frac{1}{2}$, then

$$x_5 = \frac{\pi}{3},\ x_6 = \frac{2\pi}{3},\ x_7 = \frac{4\pi}{3},\ x_8 = \frac{5\pi}{3}$$

Hence there are eight solutions

$z_k = \cos x_k + i \sin x_k,\ k = 1, 2, ..., 8$

Alternatively :

$$|z| = 1 \Rightarrow z = \frac{1}{\bar{z}}$$

hence $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1;\ z = e^{i\theta}$

$$\left|e^{i2\theta} + e^{-i2\theta}\right| = 1$$

$$|2\cos 2\theta| = 1$$

$$\cos 2\theta = \frac{1}{2} \text{ or } -\frac{1}{2}$$

16. Number of ordered pairs(s) (a, b) of real numbers such that $(a + ib)^{2008} = a - ib$ holds good, is :

युग्मों (a, b) की संख्या जिसके लिए $(a + ib)^{2008} = a - ib$ सत्य है, है :

(A) 2008 (B) 2009 (C) 2010 (D) 1

Ans. C

Sol. Let $z = a + ib \Rightarrow \bar{z} = a - ib$

hence we have $z^{2008} = \bar{z}$

$\therefore\ |z|^{2008} = |\bar{z}| = |z|$

$$|z|\left[|z|^{2007} - 1\right] = 0$$

$|z| = 0$ or $|z| = 1$; if $|z| = 0 \Rightarrow z = 0 \Rightarrow (0, 0)$

if $|z| = 1$ $z^{2009} = z\bar{z} = |z|^2 = 1 \Rightarrow$ 2009 values of z $\Rightarrow$ Total = 2010

17. If $z=\dfrac{(1+i)\sqrt{3}-(1-i)}{2\sqrt{2}}$, then :

(A) $|z| = 1$, amp $(z) = -\pi/4$ (B) $|z| = 1$, amp$(z) = \pi/4$

(C) $|z| = 1$, amp$(z) = 5\pi/12$ (D) $|z| = 1$, amp $(z) = \pi/12$

यदि $z=\dfrac{(1+i)\sqrt{3}-(1-i)}{2\sqrt{2}}$, हो, तो :

(A) $|z| = 1$, कोणांक $(z) = -\pi/4$ (B) $|z| = 1$, कोणांक $(z) = \pi/4$

(C) $|z| = 1$, कोणांक $(z) = 5\pi/12$ (D) $|z| = 1$, कोणांक $(z) = \pi/12$

Ans. C

Sol. $z=\dfrac{\left(\sqrt{3}-1\right)}{2\sqrt{2}}+\dfrac{i\left(\sqrt{3}+1\right)}{2\sqrt{2}}$

$$|z|=\sqrt{\frac{3+1-2\sqrt{3}+3+1+2\sqrt{3}}{8}}$$

$= 1$

$$\text{amp}(2)=\tan^{-1}\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)=\frac{\pi}{12}$$

18. If z_1, z_2 are two complex numbers such that $|z_1 + z_2| = |z_1 - z_2|$ then amp(z_1) –amp(z_2) is equal to :

यदि z_1, z_2 ऐसी दो सम्मिश्र संख्याएँ हैं कि $|z_1 + z_2| = |z_1 - z_2|$ तो कोणांक (z_1) –कोणांक(z_2) बराबर है :

(A) $\pi/3$ (B) $\pi/2$ (C) $\pi/4$ (D) 0

Ans. B

Sol. Let $z_1=(x_1+iy_1)$ and $z_2=(x_2+iy_2)$

Then, $(z_1+z_2)=(x_1+x_2)+i(y_1+y_2)$

and $(z_1-z_2)=(x_1-x_2)+i(y_1-y_2)$

Now, $|z_1+z_2|=|z_1-z_2|$

$\Rightarrow |z_1+z_2|^2=|z_1-z_2|^2$ $\Rightarrow (x_1+x_2)^2+(y_1+y_2)^2=(x_1-x_2)^2+(y_1-y_2)^2$

$$\Rightarrow (x_1+x_2)^2-(x_1-x_2)^2+(y_1+y_2)^2-(y_1-y_2)^2=0$$

$$\Rightarrow 4(x_1x_2+y_1y_2)=0 \qquad \Rightarrow (x_1x_2+y_1y_2)=0$$

$$\therefore \text{amp}(z_1)-\text{amp}(z_2)=\tan^{-1}\frac{y_1}{x_1}-\tan^{-1}\frac{y_2}{x_2}$$

$$=\tan^{-1}\left\{\frac{\left(\frac{y_1}{x_1}-\frac{y_2}{x_2}\right)}{1+\left(\frac{y_1}{x_1}\times\frac{y_2}{x_2}\right)}\right\}=\tan^{-1}\left\{\frac{(x_2y_1-x_1y_2)}{(x_1x_2+y_1y_2)}\right\}$$

$$=\tan^{-1}\infty=\frac{\pi}{2}\left[\because x_1x_2+y_1y_2=0\right]$$

19. The amplitude of complex number $(1+i\sqrt{3})(1+i)(\cos\theta+i\sin\theta)$ is :

सम्मिश्र संख्या $(1+i\sqrt{3})(1+i)(\cos\theta+i\sin\theta)$ का कोणांक है :

(A) $\frac{\pi}{12}-\theta$ (B) $\frac{7\pi}{12}+\theta$ (C) $\frac{7\pi}{12}-\theta$ (D) $\frac{\pi}{12}+\theta$

Ans. B

Sol. $\text{Amplitude}=\frac{\pi}{3}+\frac{\pi}{4}+\theta$

$$=\frac{7\pi}{12}+\theta$$

20. The curve represented by $\text{Re}(z^2)=4$ is :

(A) a parabola (B) an ellipse (C) a circle (D) a rectangular hyperbola

$\text{Re}(z^2)=4$ के द्वारा प्रदर्शित वक्र है :

(A) परवलय (B) दीर्घवृत्त (C) वृत्त (D) आयतीय अतिपरवलय

Ans. D

Sol. Let $z=x+iy$

$$\Rightarrow z^2=(x+iy)^2=x^2+(iy)^2+2(x)(iy)=x^2-y$$

Given $\text{Re}(z^2)=4$

$\Rightarrow x^2 - y^2 = 4$ which is rectangular hyperbola

21. The complex number $z = x + iy$ which satisfy the equation $\left|\frac{z-5i}{z+5i}\right| = 1$ lie on :

(A) the x-axis (B) the straight line $y=5$

(C) a circle passing through the origin (D) the y-axis

समीकरण $\left|\frac{z-5i}{z+5i}\right| = 1$ को संतुष्ट करेने वाली सम्मिश्र संख्या $z = x + iy$ जिस वक्र पर स्थित है, वह है :

(A) x-अक्ष (B) सरल रेखा $y=5$

(C) मूल बिन्दु से गुजरने वाला वृत्त (D) y-अक्ष

Ans. A

Sol. $|z - 5i| = |z + 5i|$ implies perpendicular bisector of line joining $5i$ & $-5i$ which is real axis.
$|z - 5i| = |z + 5i|$ का अर्थ $5i$ और $-5i$ को मिलाने वाली रेखा के लम्बवत् समद्विभाजक है जो वास्तविक अक्ष है।

22. Locate the points representing the complex number z on the Argand plane :

सम्मिश्र तल में सम्मिश्र संख्या z द्वारा प्रदर्शित बिन्दुओं को दर्शाइये :

(a) $|z+1-2i| = \sqrt{7}$ (b) $|z-1|^2 + |z+1|^2 = 4$

(c) $\left|\frac{z-3}{z+3}\right| = 3$; (d) $|z-3| = |z-6|$

Ans. **(a)** on a circle of radius $\sqrt{7}$ with centre $(-1, 2)$**;** **(b)** on a unit circle with centre at origin
(c) on a circle with centre $(-15/4, 0)$ and radius $9/4$**;** **(d)** a straight line

Sol.

23. Find the modulus, argument and the principal argument of the complex numbers :

सम्मिश्र संख्याओं का मापांक, कोणांक तथा मुख्य कोणांक ज्ञात कीजिये :

(i) $6(\cos 310° - i \sin 310°)$ (ii) $-2(\cos 30° + i \sin 30°)$

(iii) $\frac{2+i}{4i + (1+i)^2}$

Ans. (i) Modulus $= 6$, Arg $= 2k\pi + \frac{5\pi}{18}$ $(K \in I)$, Principal Arg $= \frac{5\pi}{18}$ $(K \in I)$

(ii) Modulus $= 2$, Arg $= 2k\pi + \frac{7\pi}{6}$, Principal Arg $= -\frac{5\pi}{6}$

(iii) $\text{Modulus} = \frac{\sqrt{5}}{6}$, $\text{Arg} = 2k\pi - \tan^{-1}2\ (K \in I)$, $\text{Principal Arg} = -\tan^{-1}2$

Sol.

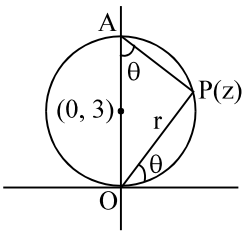
1. Let z be a complex number having the argument θ, $0 < \theta < \pi/2$ and satisfying the equality $|z-3i| = 3$. Then $\cot\theta - \frac{6}{z}$ is equal to :

यदि z एक सम्मिश्र संख्या है, जिसका कोणांक θ, $0 < \theta < \pi/2$, $|z-3i| = 3$ को सन्तुष्ट करता है। तब $\cot\theta - \frac{6}{z}$ बराबर :

(A) 1 (B) -1 (C) i (D) $-i$

Ans. C

Sol. $z = r(\cos\theta + i\sin\theta)$ now $r = OA\sin\theta = 6\sin\theta$



$z = 6\sin\theta(\cos\theta + i\sin\theta)$ $\quad \frac{6}{z} = \frac{1}{\sin\theta(\cos\theta + i\sin\theta)}$

$= \frac{\cos\theta - i\sin\theta}{\sin\theta} = -i + \cot\theta \Rightarrow \cot\theta - \frac{6}{z} = i \Rightarrow$ C

2. If the complex number z satisfies the condition $|z| \geq 3$, then the least value of $\left|z + \frac{1}{z}\right|$ is equal to :

(A) 5/3 (B) 8/3 (C) 11/3 (D) None of these

यदि सम्मिश्र संख्या z प्रतिबन्ध $|z| \geq 3$ को सन्तुष्ट करता है तब $\left|z + \frac{1}{z}\right|$ का न्यूनतम मान है :

(A) 5/3 (B) 8/3 (C) 11/3 (D) इनमें से कोई नहीं

Ans. B

Sol. $\left|z + \frac{1}{z}\right| \geq |z| - \frac{1}{|z|}$

$\left|z + \frac{1}{z}\right|_{least} \geq 3 - \frac{1}{3} \geq \frac{8}{3}$

3. Given $z_p = \cos\left(\frac{\pi}{2^P}\right) + i\sin\left(\frac{\pi}{2^P}\right)$, then $\lim_{n\to\infty}(z_1 z_2 z_3 .... z_n) =$

दिया है $z_p = \cos\left(\frac{\pi}{2^P}\right) + i\sin\left(\frac{\pi}{2^P}\right)$, तब $\lim_{n\to\infty}(z_1 z_2 z_3 .... z_n) =$

(A) 1 (B) -1 (C) i (D) $-i$

Ans. B

Sol. $z_p = e^{\frac{i\pi}{2^p}}$; $z_1 = e^{\frac{i\pi}{2}}$; $z_2 = e^{\frac{i\pi}{2^2}}$ and so on .....

$$\lim_{n\to\infty} z_1 z_2 .....z_n = \lim_{n\to\infty} e^{i\pi\left(\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+.....+\frac{1}{2^n}\right)}$$

$$= e^{i\pi}\left(\frac{1/2}{1-1/2}\right) = e^{i\pi} = \cos\pi + i\sin\pi = -1$$

4. The maximum and minimum values of $|z+1|$ when $|z+3| \le 3$ are :

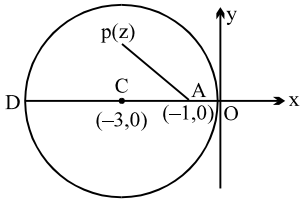
तब $|z+1|$ का अधिकतम और न्यूनतम मान है यदि $|z+3| \le 3$:

(A) $(5,0)$ (B) $(6,0)$ (C) $(7,1)$ (D) $(5,1)$

Ans. A

Sol. $|z+3| \le 3$ denotes set of points
on or inside a circle with centre $(-3, 0)$ and

radius 3 . $|z+1|$ denotes the distance of P



from A of $|z+1|_{min} = 0$ &
$|z+1|_{max} = AD$

5. If z_1, z_2, z_3 are the vertices of the ΔABC on the complex plane which are also the roots of the equation, $z^3 - 3\alpha z^2 + 3\beta z + x = 0$, then the condition for the ΔABC to be equilateral triangle is :

यदि $z_1, z_2, z_3, \Delta ABC$ के शीर्ष है तथा $z^3 - 3\alpha z^2 + 3\beta z + x = 0$ के मूल भी है तब ΔABC के समबाहु होने का प्रतिबन्ध होगा :

(A) $\alpha^2 = \beta$ (B) $\alpha = \beta^2$ (C) $\alpha^2 = 3\beta$ (D) $\alpha = 3\beta^2$

Ans. A

Sol. $z_1 + z_2 + z_3 = 3\alpha$; $\sum z_1 z_2 = 3\beta$

If ABC is equilateral $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$

$$(z_1 + z_2 + z_3)^2 = 3\sum z_1 z_2$$

$9\alpha^2 = 3 \cdot 3\beta = 9\beta \Rightarrow \alpha^2 = \beta$

6. Let ABCDEF be a regular hexagon with vertices in the anticlockwise sense. If z_1 and z_2 represent A and B respectively, the complex number representing C is : (Where $\omega = e^{i2\pi/3}$)

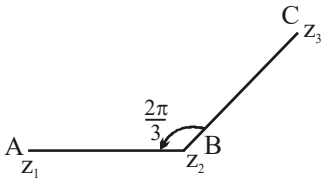
माना ABCDEF एक समबाहु षट्भुज है जिसके शीर्ष वामावर्त लिये गये है। यदि z_1 एवं z_2 क्रमशः A तथा B है, तो सम्मिश्र संख्या जो C को निरूपित करती है, होगी : (जहाँ $\omega = e^{i2\pi/3}$)

(A) $\omega z_1 + (1 + \omega) z_2$ (B) $\omega^2 z_1 + (1 - \omega^2) z_2$ (C) $\omega z_1 + (\omega - 1) z_2$ (D) $\omega z_1 + (1 - \omega) z_2$

Ans. B

Sol. Rotation about B gives [V-group 2009]

$z_1 - z_2 = (z_3 - z_2)\, w$



$\Rightarrow z_3 - z_2 = w^2 (z_1 - z_2)$

$\Rightarrow z_3 = w^2 z_1 + (1 - w^2) z_2$

7. The points $z_1 = 3 + \sqrt{3}\, i$ and $z_2 = 2\sqrt{3} + 6i$ are given on a complex plane. The complex number lying on the bisector of the angle formed by the vectors z_1 and z_2 is :

(A) $z = \frac{\left(3 + 2\sqrt{3}\right)}{2} + \frac{\sqrt{3} + 2}{2} i$ (B) $z = 5 + 5i$

(C) $z = -1 - i$ (D) None

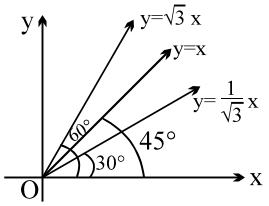
यदि $z_1 = 3 + \sqrt{3}\, i$ और $z_2 = 2\sqrt{3} + 6i$ सम्मिश्र तल में कोई दो बिन्दु दिये है। तब z_1 और z_2 के मध्य कोणार्धक पर सम्मिश्र संख्या होगी :

(A) $z = \frac{\left(3 + 2\sqrt{3}\right)}{2} + \frac{\sqrt{3} + 2}{2} i$ (B) $z = 5 + 5i$

(C) $z = -1 - i$ (D) कोई नहीं

Ans. B

Sol. Note that $z_1 = 3 + \sqrt{3}$ i lies on the line $y = \frac{1}{\sqrt{3}} x$ & $z_2 = 2\sqrt{3} + 6$ i lies on the line $y = \sqrt{3} x$. Hence $z = 5 + 5$ i



will only lie on the bisector of z_1 & z_2 i.e. $y = x$

8. Let z_1 and z_2 be non zero complex numbers satisfying the equation, $z_1^2 - 2z_1z_2 + 2z_2^2 = 0$. The geometrical nature of the triangle whose vertices are the origin and the points representing z_1 and z_2 is :

(A) an isosceles right angled triangle

(B) a right angled triangle which is not isosceles

(C) an equilateral triangle

(D) an isosceles triangle which is not right angled

माना z_1 और z_2 अशून्य सम्मिश्र संख्या, समीकरण $z_1^2 - 2z_1z_2 + 2z_2^2 = 0$ को सन्तुष्ट करते है। तब त्रिभुज जिनके शीर्ष मूल बिन्दु और z_1, z_2 है, की ज्यामिती प्रकृति होगी :

(A) समद्विबाहु समकोण त्रिभुज

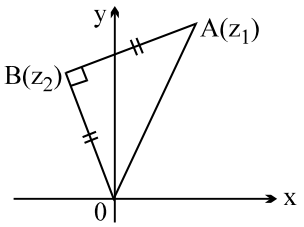
(B) समकोण लेकिन समद्विबाहु नहीं

(C) समबाहु त्रिभुज

(D) समद्विबाहु त्रिभुज जो समकोण नहीं है

Ans. A

Sol. $\frac{z_1}{z_2} = z \Rightarrow z^2 - 2z + 2 = 0 \Rightarrow z = 1 \pm i$

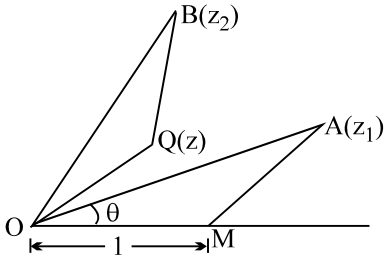


$\Rightarrow \frac{z_1}{z_2} = 1 \pm i \Rightarrow z_1 = z_2 \pm z_2 i \Rightarrow z_1 - z_2 = \pm z_2 i$

$\Rightarrow$ $z_1 - z_2$ is perpendicular to z_2 and $|z_1 - z_2| = |z_2|$

9. On the Argand plane point 'A' denotes a complex number z_1. A triangle OBQ is made directily similiar to the triangle OAM, where OM = 1 as shown in the figure. If the point B denotes the complex number z_2, then the complex number corresponding to the point 'Q' is :

सम्मिश्र तल में बिन्दु A, z_1 है तथा त्रिभुज OBQ $\cong$ OAM, जहां OM = 1 जैसा चित्र में दर्शाया गया है। यदि B, z_2 है तब बिन्दु 'Q ' सम्मिश्र संख्या दर्शाता है :



(A) $z_1 z_2$ (B) $\frac{z_1}{z_2}$ (C) $\frac{z_2}{z_1}$ (D) $\frac{z_1 + z_2}{z_2}$

Ans. C

Sol. $\frac{z_2}{|z_2|} = \frac{z}{|z|} e^{i\theta}$ .....(1); $\frac{z_1}{|z_1|} = 1\ e^{i\theta}$ ....(2)

substitute the value of $e^{i\theta}$ from (2) in (1)

$\frac{z}{|z|} = \frac{z_2}{|z_2|} \cdot \frac{|z_1|}{z_1} \Rightarrow \frac{z}{|z|} = \frac{z_2/z_1}{|z_2/z_1|}$; $z = \frac{z_2}{z_1}$

10. If $i = \sqrt{-1}$, then $4 + 5\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{334} + 3\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{365}$ is equal to :

यदि $i = \sqrt{-1}$, तब $4 + 5\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{334} + 3\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{365}$ बराबर :

(A) $1 - i\sqrt{3}$ (B) $-1 + i\sqrt{3}$ (C) $i\sqrt{3}$ (D) $-i\sqrt{3}$

Ans. C

Sol. $E = 4 + 5\omega^{334} + 3\omega^{365} = 4 + 5\omega + 3\omega^2 = 3(1 + \omega + \omega^2) + 1 + 2\omega = (1 + \omega) + \omega = \omega - \omega^2$

11. If ω is imaginary cube root of unity, then the value of $\frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2} + \frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2}$ is :

यदि ω इकाई का काल्पनिक घनमूल हो, तो $\frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2} + \frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2}$ का मान होगा :

(A) 1 (B) 0 (C) – 1 (D) 2

Ans. C

Sol. $\frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2} + \frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2}$

$$\Rightarrow \frac{\omega\left(\frac{a}{\omega}+b+c\omega\right)}{b+c\omega+a\omega^2}+\frac{\omega^2\left(\frac{a}{\omega^2}+\frac{b}{\omega}+c\right)}{c+a\omega+b\omega^2}$$

$$\Rightarrow \frac{\omega\left(a\omega^2+b+c\omega\right)}{\left(b+c\omega+a\omega^2\right)}+\frac{\omega^2\left(a\omega+b\omega^2+c\right)}{c+a\omega+b\omega^2}$$

$$\Rightarrow \omega+\omega^2=-1$$

12. If ω is imaginary cube root of unity and if $n = 3k + 2$ then the value of $\omega^n + \omega^{2n}$ is :

यदि ω इकाई का काल्पनिक घनमूल हो तथा यदि $n = 3k + 2$ हो, तो $\omega^n + \omega^{2n}$ का मान होगा :

(A) 0 (B) –1 (C) 2 (D) 1

Ans. B

Sol. $n = 3k + 2$

$\omega^n + \omega^{2n} = \omega^{3k+2} + \omega^{6k+4}$

$= \omega^{3k}.\omega^2 + \omega^{6k}.\omega^4$

$= \omega^2 + \omega^4$

$= \omega^2 + \omega$

$= -1$

13. If ω is imaginary cube root of unity then the value of $(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8)$..... 2n is :

यदि ω इकाई का काल्पनिक घनमूल हो, तो $(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8)$ ..... 2n गुणनखण्ड तक मान होगा :

(A) 0 (B) n (C) –1 (D) 1

Ans. D

Sol. Considering first 2 terms, we get,

$(1 + w)(1 + w^2) = 1 + w^2 + w + w^3 = 0 + w^3 = 1$

similarly, considering second 2 terms, we get,

$\left(1+w^4\right)\left(1+w^8\right)=(1+w)\left(1+w^2\right)=1$

It continues for all the remaining pair of terms,

therefore the product upto 2n terms equal 1.

14. If ω is an imaginary cube root of unity, then for positive integral value of n, the product of $\omega.\omega^2.\omega^3.....\omega^n$ will be:

यदि ω इकाई का काल्पनिक घनमूल हो, तो n के धनात्मक पूर्णांक मानों के लिये गुणनफल $\omega.\omega^2.\omega^3.....\omega^n$ होगा :

(A) 1 (B) $\frac{1-i\sqrt{3}}{2}$ (C) $\frac{1+i\sqrt{3}}{2}$ (D) 1, ω

Ans. D

Sol. $\Rightarrow \omega \times \omega^2 \times \omega^3 ..... \times \omega^n$

$\Rightarrow \omega^{\frac{n(n+1)}{2}}$

Case – 1

$n \rightarrow$ even

$n + 1 \rightarrow$ odd

$\omega^{odd} \rightarrow 1, \omega$

Case – 2

$n \rightarrow$ odd

$n + 1 \rightarrow$ even

$\omega^{odd} \rightarrow 1, \omega$

15. If $|z| = 2$, then locus of $-1 + 5z$ is a circle whose centre is :

यदि $|z| = 2$, तो $-1 + 5z$ का बिन्दुपथ एक वृत्त होगा जिसका केन्द्र है :

(A) (–1, 0) (B) (1, 0) (C) (0, –1) (D) (0, 0)

Ans. A

Sol. $z' = -1 + 5z$

$z' = -1 + 5z$

$|z' + 1| = |5z|$

$|z' + 1| = 5|z| = 5(2) = 10$

$|x + iy + 1| = 10$

$\Rightarrow (x+1)^2 + y^2 = 100$

Which is an equation of circle with raidus 10 and center (–1, 0)

1. z is a complex number such that $z + \frac{1}{z} = 2\cos 3°$, then the value of $z^{2000} + \frac{1}{z^{2000}} + 1$ is equal to :

   यदि z सम्मिश्र संख्या है इस प्रकार कि $z + \frac{1}{z} = 2\cos 3°$, तब $z^{2000} + \frac{1}{z^{2000}} + 1$ का मान है :

   (A) 0 (B) – 1 (C) $\sqrt{3}+1$ (D) $1 - \sqrt{3}$

Ans. A

Sol. Let $z = \cos\theta + i\sin\theta = = e^{i\theta}$; $\frac{1}{z} = \cos\theta - i\sin\theta = e^{-i\theta}$

so that $z + \frac{1}{z} = 2\cos\theta \quad (\theta = 3°)$

now $z^{2000} + \frac{1}{z^{2000}} + 1$

$e^{i\,2000\,\theta} + e^{-i\,2000\,\theta} + 1 = 2\cos(2000\,\theta) + 1 = 2\cos(6000°) + 1$ (as $\theta = 3°$)

$= 2\cos\left(\frac{100\pi}{3}\right) + 1 = 2\cos\left(\frac{4\pi}{3}\right) + 1 = -1 + 1 = 0$

2. If $|z-2| \geq |z-4|$ then correct statement is :

   यदि $|z-2| \geq |z-4|$ तो सही कथन है :

   (A) $\text{Re}(z) \geq 3$ (B) $\text{Re}(z) \leq 3$ (C) $\text{Re}(z) \geq 2$ (D) $\text{Re}(z) \leq 2$

Ans. A

Sol. Let, $z = x + iy$

Putting in the given inequality,

$\left|(x-4)-iy\right| < \left|(x-2)+iy\right|$

$\Rightarrow \sqrt{(x-4)^2 + y^2} < \sqrt{(x-2)^2 + y^2}$

squaring both sides,

$\Rightarrow (x-4)^2 + y^2 < (x-2)^2 + y^2$

$\Rightarrow -8x + 16 < -4x + 4$

$\Rightarrow 4x > 12 \Rightarrow x > 3 \Rightarrow \text{Re}(z) > 3$

3. If $z^4 + 1 = \sqrt{3}\,i$:

(A) z^3 is purely real

(B) z represents the vertices of a square of side $2^{1/4}$

(C) z^9 is purely imaginary

(D) z represents the vertices of a square of side $2^{3/4}$

यदि $z^4 + 1 = \sqrt{3}\,i$:

(A) z^3 पूर्णतः वास्तविक है

(B) z, $2^{1/4}$ भुजा वाले वर्ग के शीर्ष है

(C) z^9 पूर्णतः काल्पनिक है

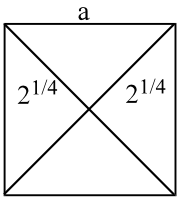
(D) z, $2^{3/4}$ भुजा वाले वर्ग के शीर्ष है

Ans. D

Sol. $z^4 = -1 + \sqrt{3}\,i = 2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$

$z^4 = 2w \Rightarrow$ A, C are not possible

root are $z_1 = 2^{1/4}\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$; $z_2 = 2^{1/4}\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$ etc.



$\therefore \quad a = \sqrt{2^{1/2} + 2^{1/2}} = (2^{3/2})^{1/2} = 2^{3/4} \Rightarrow$ (D)

4. If $z_1 = 2 + 3\,i$, $z_2 = 3 - 2\,i$ and $z_3 = -1 - 2\sqrt{3}\,i$ then which of the following is true?

यदि $z_1 = 2 + 3\,i$, $z_2 = 3 - 2\,i$ और $z_3 = -1 - 2\sqrt{3}\,i$ तब निम्न में से सही कथन है?

(A) $\arg\left(\frac{z_3}{z_2}\right) = \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)$

(B) $\arg\left(\frac{z_3}{z_2}\right) = \arg\left(\frac{z_2}{z_1}\right)$

(C) $\arg\left(\frac{z_3}{z_2}\right) = 2\arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)$

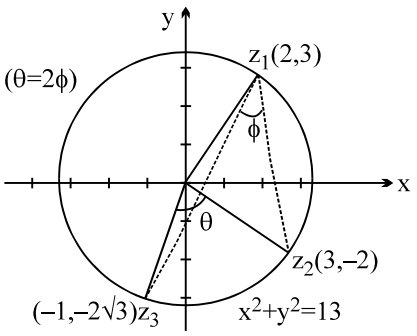
(D) $\arg\left(\frac{z_3}{z_2}\right) = \frac{1}{2}\arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)$

Ans. C

Sol. Note that $|z_1| = |z_2| = |z_3| = \sqrt{13}$

Hence z_1, z_2, z_3 lies on a circle with centre (0, 0)

and $r = \sqrt{13}$ as shown



now $\quad \text{Arg}\dfrac{z_2}{z_3} = 2\text{Arg}\dfrac{z_2 - z_1}{z_3 - z_1}$

$\therefore \quad \text{Arg}\dfrac{z_3}{z_2} = 2\text{Arg}\dfrac{z_3 - z_1}{z_2 - z_1} \quad \Rightarrow \quad$ (C)

5. If $|z^2 - 1| = |z|^2 + 1$, then z lies on :

(A) the real axis (B) the imaginary axis (C) a circle (D) an ellipse

यदि $|z^2 - 1| = |z|^2 + 1$, तो z स्थित है :

(A) वास्तविक अक्ष पर (B) काल्पनिक अक्ष पर (C) एक वृत्त पर (D) एक दीर्घवृत्त

Ans. B

Sol. Let $z = x + iy$

$z^2 - 1 = (x^2 - y^2 - 1) + 2ixy$

$|z|^2 + 1 = x^2 + y^2 + 1$

Hence

$|z^2 - 1| = |z|^2 + 1$

$(x^2 - y^2 - 1)^2 + 4x^2y^2 = (x^2 + y^2 + 1)^2$

$4x^2y^2 = (x^2 + y^2 + 1)^2 - (x^2 - y^2 - 1)^2$

$4x^2y^2 = (2x^2)(2y^2 + 2)$

$x^2y^2 = (x^2)(y^2+1)$

$x^2y^2 = x^2y^2 + x^2$

$x^2 = 0$

$x = 0$

Hence, z lies on imaginary axis

6. If z is a complex number satisfying the equation $Z^6 + Z^3 + 1 = 0$. If this equation has a root $re^{i\theta}$ with $90° < \theta < 180°$ then the value of 'θ' is :

यदि z सम्मिश्र संख्या समीकरण $Z^6 + Z^3 + 1 = 0$ को सन्तुष्ट करती है। यदि इसे समीकरण का मूल $re^{i\theta}$ $(90° < \theta < 180°)$ है तब 'θ' का मान होगा :

(A) 100° (B) 110° (C) 160° (D) 170°

Ans. C

Sol. Let $Z^3 = t$

hence equation becomes

$t^2 + t + 1 = 0 \Rightarrow t = \omega$ or ω^2

$$Z^3 = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3} = e^{\left(2m\pi+\frac{2\pi}{3}\right)i}$$

$$Z = e^{\frac{\left(2m\pi+\frac{2\pi}{3}\right)}{3}i}$$

put $m = 1$ to get $\theta = \frac{8\pi}{9} \in (90°, 180°) = 160°$

7. If $w = \frac{z}{z - \frac{1}{3}i}$ and $|w| = 1$, then z lies on :

(A) an ellipse (B) a circle (C) a straight line (D) a parabola

यदि $w = \frac{z}{z - \frac{1}{3}i}$ तथा $|w| = 1$, तब z स्थित है :

(A) एक दीर्घ वृत्त पर (B) एक वृत्त पर (C) एक रेखा पर (D) एक परवलय पर

Ans. C

Sol. Given that $\omega = \dfrac{z}{z-\dfrac{i}{3}}$ and $|\omega| = 1$

$$\therefore \quad |\omega| = \left|\frac{z}{z-\frac{i}{3}}\right| \Rightarrow \frac{|z|}{\left|z-\frac{i}{3}\right|} = 1$$

$$\Rightarrow |z| = \left|z-\frac{i}{3}\right| \Rightarrow -\frac{2}{3}y + \frac{1}{9} = 0$$

Which is a straight line.

8. P(z) is the point moving in the Argand's plane satisfying $\arg(z-1) - \arg(z+i) = \pi$ then, P is :

(A) a real number, hence lies on the real axis

(B) an imaginary number, hence lies on the imaginary axis

(C) a point on the hypotenuse of the right angled triangle OAB formed by O ≡ (0, 0); A ≡ (1, 0); B ≡ (0, – 1)

(D) a point on an arc of the circle passing through A ≡ (1, 0); B ≡ (0, – 1)

यदि बिन्दु P(z) सम्मिश्र तल $\arg(z-1) - \arg(z+i) = \pi$ को सन्तुष्ट करता है, तब P है :

(A) एक पूर्णतः वास्तविक संख्या, अतः वास्तविक अक्ष पर होगा

(B) एक पूर्णतः काल्पनिक संख्या अतः काल्पनिक अक्ष पर होगा

(C) बिन्दुओं O ≡ (0, 0); A ≡ (1, 0); B ≡ (0, – 1) से निर्मित समकोण त्रिभुज के कर्ण पर बिन्दु है

(D) बिन्दुओं A ≡ (1, 0); B ≡ (0, – 1) से जाने वाले वृत्त के चाप पर कोई बिन्दु

Ans. C

Sol. $\text{amp.}\left(\dfrac{z-1}{z+i}\right) = \pi \Rightarrow \dfrac{z-1}{z+i}$ is real $\Rightarrow$ z moves on the lines joining (0, – 1) and (1, 0)

9. If z_1 and z_2 are two complex numbers & if $\arg\left(\dfrac{z_1+z_2}{z_1-z_2}\right) = \dfrac{\pi}{2}$ but $|z_1 + z_2| \neq |z_1 - z_2|$ then the figure formed by the points represented by $0, z_1, z_2$ and $z_1 + z_2$ is :

(A) a parallelogram but not a rectangle or a rhombous

(B) a rectangle but not a square

(C) a rhombous but not a square

(D) a square

यदि z_1 और z_2 कोई दो सम्मिश्र संख्याऐं हैं, यदि $\arg\left(\frac{z_1+z_2}{z_1-z_2}\right)=\frac{\pi}{2}$ लेकिन $|z_1+z_2| \neq |z_1-z_2|$ तब बिन्दुओं $0, z_1, z_2$ और z_1+z_2 द्वारा निर्मित आकृति है :

(A) समानान्तर चतुर्भुज लेकिन आयत और समचर्तुभुज नहीं

(B) आयत लेकिन वर्ग नहीं

(C) समचतुर्भुज लेकिन वर्ग नहीं

(D) एक वर्ग

Ans. C

Sol. $\arg\left(\frac{z_1+z_2}{z_1-z_2}\right)=\frac{\pi}{2}$

$|z_1+z_2| \neq |z_1-z_2|$

$\therefore$ Figure formed by $0, z_1z_2, z_1+z_2$

$\Rightarrow \arg(z_1+z_2)-\arg(z_1-z_2)=\frac{\pi}{2}$

$\therefore \arg(z_1+z_2)=\frac{\pi}{2}+\arg(z_1-z_2)$

$\Rightarrow z_1=r_1e^{iQ_1} z_2=r_2e^{iQ_2} z_3=r_3e^{iQ_3}$

$z_4=r_4e^{iQ_4}$

$\Rightarrow z_1+z_2=r_1e^{iQ_1}+r_2e^{iQ_2}$

$\Rightarrow z_1-z_2=r_1e^{iQ_1}-r_2e^{iQ_2}$

$\therefore$ The points represent a rhombus.

Hence, the answer is a rhombus but not a square.

10. If A and B be two complex numbers satisfying $\frac{A}{B}+\frac{B}{A}=1$. Then the two points represented by A and B and the origin form the vertices of :

(A) an equilateral triangle

(B) an isosceles triangle which is not equilateral

(C) an isosceles triangle which is not right angled

(D) a right angled triangle

यदि A और B कोई दो सम्मिश्र संख्याऐं इस प्रकार है $\frac{A}{B}+\frac{B}{A}=1$. तब बिन्दुओं A, B और मूल बिन्दु किसके शीर्ष होंगे :

(A) एक समबाहु त्रिभुज

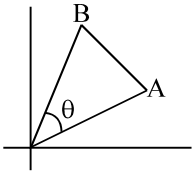
(B) समद्विबाहु जो समबाहु नहीं

(C) समद्विबाहु त्रिभुज जो समकोण नहीं है

(D) एक समकोण त्रिभुज

Ans. A

Sol. $A^2 - AB + B^2 = 0$



Let $\frac{A}{B}=z;\ \Rightarrow z+\frac{1}{z}=1$ hence $z^2 - z + 1 = 0 \Rightarrow z = -\omega$ or $-\omega^2$

$\Rightarrow$ $A = -B\omega$ or $A = -B\omega^2 \Rightarrow |A| = |B|$

and amp (A) – amp (B) = amp $(-\omega)$ = amp (-1) + amp $(\omega) = \pi + \frac{2\pi}{3}$

11. If $1, \alpha_1, \alpha_2 ....., \alpha_{2008}$ are $(2009)^{th}$ roots of unity, then the value of $\sum_{r=1}^{2008} r(\alpha_r + \alpha_{2009-r})$ equals :

यदि $1, \alpha_1, \alpha_2 ....., \alpha_{2008}$ इकाई के $(2009)^{th}$ मूल है तब $\sum_{r=1}^{2008} r(\alpha_r + \alpha_{2009-r})$ बराबर :

(A) 2009 (B) 2008 (C) 0 (D) – 2009

Ans. D

Sol. Let $S = 1(\alpha_1 + \alpha_{2008}) + 2(\alpha_2 + \alpha_{2007}) + 3(\alpha_3 + \alpha_{2006}) + ... + 2008(\alpha_{2008} + \alpha_1)$ ....(1)

Also $S = 2008(\alpha_{2008} + \alpha_1) + 2007(\alpha_2 + \alpha_{2007}) + ... + 2(\alpha_2 + \alpha_{2007}) + 1(\alpha_1 + \alpha_{2008})$ ....(2)

(writing in reverse order)

$\therefore$ On adding (1) and (2), we get

$$2S = 2009[2(\alpha_1 + \alpha_2 + \alpha_3 + ...\alpha_{2008})]$$
$$2S = 2009[2(\underbrace{1 + \alpha_1 + \alpha_2 + \alpha_3 + ...\alpha_{2008}}_{\text{zero}} - 1)]$$

Hence $S = -2009$ Ans.

Note that $(\alpha_1$ and $\alpha_{2008})$, $(\alpha_2$ and $\alpha_{2007})$, $(\alpha_3$ and $\alpha_{2006})$, ..., $(\alpha_{1004}$ and $\alpha_{1005})$ are conjugate of each other.

12. The value of $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7} - i\cos\frac{2\pi k}{7}\right)$ is :

$\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7}-i\cos\frac{2\pi k}{7}\right)$ का मान है :

(A) -1 (B) 0 (C) $-i$ (D) i

Ans. D

Sol. $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7}-i\cos\frac{2\pi k}{7}\right)$

$=\frac{1}{i}\sum_{k=1}^{6}\left(\cos\frac{2\pi k}{7}+i\sin\frac{2\pi k}{7}\right)$

$=\frac{1}{i}\left(0-\cos\frac{14\pi}{7}+i\left(0-\sin\frac{14\pi}{7}\right)\right)$

$=i$

13. If $\alpha_1,\ \alpha_2,\ \alpha_3,...\ \alpha_8$ are the 8, 8th roots of unity then :

यदि $\alpha_1,\ \alpha_2,\ \alpha_3,...\ \alpha_8$ इकाई के 8, 8वें मूल हों, तो :

(A) $(\alpha_1)^3+(\alpha_2)^3+(\alpha_3)^3+...+(\alpha_8)^3=8$

(B) $\alpha_1+\alpha_2+\alpha_3+...+\alpha_8=1$

(C) $(\alpha_1)^{50}+(\alpha_2)^{50}...+(\alpha_8)^{50}=0$

(D) $(\alpha_1)^{16}+(\alpha_2)^{16}...+(\alpha_8)^{16}=0$

Ans. C

Sol.

14. A polynomial f(z) when divided by $(z-w)$ leaves remainder $2+i\sqrt{3}$ and when divided by $(z-w^2)$ leaves remainder $2-i\sqrt{3}$. If the remainder obtained when f(z) is divided by z^2+z+1 is $az+b$ (where w is a non- real cube root of unity and $a, b \in R^+$), then find the value of $(a+b)$.

एक बहुपद f(z) को $(z-w)$ से भाग देने पर शेषफल $2+i\sqrt{3}$ होता है, तथा जब $(z-w^2)$ से भाग दे तो शेषफल $2-i\sqrt{3}$ होता है। यदि f(z) को z^2+z+1 से भाग देने पर शेषफल $az+b$ है (जहाँ w, इकाई का काल्पनिक घनमूल है एवं $a, b \in R^+$), तो $(a+b)$ का मान ज्ञात करें।

Ans. 0005

Sol. We have $f(w)=2+i\sqrt{3}$, $f(w^2)=2-i\sqrt{3}$

Let $f(z)=(z^2+z+1)g(z)+(az+b)$

$\therefore$ $aw+b=2+i\sqrt{3}$ ....(1)

and $aw^2 + b = 2 - i\sqrt{3}$ ....(2)

$\therefore$ On solving (1) and (2), we get

$a = 2,\ b = 3 \quad \Rightarrow \quad a + b = 5$

1. If the six solutions of $x^6 = -64$ are written in the form $a + bi$, where a and b are real, then the product of those solutions with $a > 0$, is :

   यदि $x^6 = -64$ के सभी हलों को $a + bi$, के रूप में लिखाा जा सकता है जहां a और b वास्तविक संख्या है तब हलों का गुणनफल (जहां $a > 0$) है :

   (A) 4 (B) 8 (C) 16 (D) 64

Ans. A

Sol. Use De Moivre's theorem get the product of roots with + ve real part

$$z = 2(-1)^{1/6} = 2\left[\cos\frac{(2m+1)\pi}{6} + i\sin\frac{(2m+1)\pi}{6}\right]$$

put $m = 0$ or $m = 5$ for positive real part to get

$z_1 z_2 = 4e^{i\pi/6} \cdot e^{(11\pi/6)i} = 4e^{2\pi i} = 4$

2. If $z_n = \cos\frac{\pi}{(2n+1)(2n+3)} + i\sin\frac{\pi}{(2n+1)(2n+3)}$, then $\underset{n\to\infty}{\text{Limit}}\,(z_1 . z_2 . z_3 . ..... z_n) =$

   यदि $z_n = \cos\frac{\pi}{(2n+1)(2n+3)} + i\sin\frac{\pi}{(2n+1)(2n+3)}$, तब $\underset{n\to\infty}{\text{Limit}}\,(z_1 . z_2 . z_3 . ..... z_n) =$

   (A) $\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$ (B) $\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}$ (C) $\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}$ (D) $\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}$

Ans. B

Sol. $z_n = e^{\frac{i\pi}{2}\left(\frac{1}{2n+1} - \frac{1}{2n+3}\right)}$

$$\prod_{n=1}^{\infty} z_n = e^{\frac{i\pi}{2}\left[\left(\frac{1}{3}-\frac{1}{5}\right)+\left(\frac{1}{5}-\frac{1}{7}\right)+.....+\left(\frac{1}{2n+1}-\frac{1}{2n+3}\right)\right]} = e^{\frac{i\pi}{2}\left[\left(\frac{1}{3}-\frac{1}{2n+3}\right)\right]} = e^{\frac{i\pi}{6}} \text{ as } n\to\infty \Rightarrow \text{(B)}$$

3. The straight line $(1+2i)z + (2i-1)\bar{z} = 10i$ on the complex plane, has intercept on the imaginary axis equal to :

   सम्मिश्र तल में सरल रेखा $(1 + 2i)z + (2i - 1)\bar{z} = 10i$ का काल्पनिक अक्ष पर अन्तःखण्ड है :

   (A) 5 (B) $\frac{5}{2}$ (C) $-\frac{5}{2}$ (D) -5

Ans. A

Sol. put $z = iy$ $\quad (1+2i)\,iy - (2i-1)\,i\,y = 10\,i$

$2y + 0\,y = 10 \;\Rightarrow\; y = 5$

Note: For x-intercept put $z = x + 0i \;\Rightarrow\; x = 5/2$

Alternatively: put $z + \bar{z} = 0 \;\Rightarrow\; \bar{z} = -z \;\Rightarrow\; (1+2i)z - z(2i-1) = 10i$

$2z = 10i \;\Rightarrow\; z = 5i\,; \quad y = 5$

4. Intercept made by the circle $z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + r = 0$ on the real axis on complex plane, is :

वह $z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + r = 0$ का वास्तविक अक्ष पर अन्तःखण्ड है :

(A) $\sqrt{(\alpha+\bar{\alpha})-r}$ (B) $\sqrt{(\alpha+\bar{\alpha})^2-2r}$ (C) $\sqrt{(\alpha+\bar{\alpha})^2+r}$ (D) $\sqrt{(\alpha+\bar{\alpha})^2-4r}$

Ans. D

Sol. Points where the circle cuts the x-axis $z = \bar{z}$.

Hence substituting $z = \bar{z}$ in the equation of circle

$z^2 + \bar{\alpha}z + \alpha z + r = 0$

$z^2 + (\alpha+\bar{\alpha})z + r = 0$

$AB = |z_1 - z_2| = \sqrt{(z_1+z_2)^2 - 4z_1z_2} = \sqrt{(\alpha+\bar{\alpha})^2 - 4r} \;\Rightarrow$ (D)

Alternatively: put $z = x$ and $\bar{z} = x$ to get $x^2 + \bar{\alpha}x + \alpha x + r = 0$ which is the same equation

5. If Z_r ; $r = 1, 2, 3,..., 50$ are the roots of the equation $\sum_{r=0}^{50} (Z)^r = 0$, then the value of $\sum_{r=1}^{50} \frac{1}{Z_r - 1}$ is :

यदि Z_r ; $r = 1, 2, 3,..., 50$ समीकरण $\sum_{r=0}^{50} (Z)^r = 0$, के मूल है, तब $\sum_{r=1}^{50} \frac{1}{Z_r - 1}$ का मान है :

(A) -85 (B) -25 (C) 25 (D) 75

Ans. B

Sol. $E = \frac{1}{z_1 - 1} + \frac{1}{z_2 - 1} + ... + \frac{1}{z_{50} - 1}$, where $z_1, z_2,...., z_{50}$ are the roots of the equation $z^{51} - 1 = 0$ other than 1.

$= -25 + \left(\frac{1}{2} + \frac{1}{z_1 - 1}\right) + \left(\frac{1}{2} + \frac{1}{z_2 - 1}\right) + ... + \left(\frac{1}{2} + \frac{1}{z_{50} - 1}\right)$

Note that (1st + last) and (2nd + 2nd last) will vanish using $z_r = z^r$ and $z^{51} = 1$

6. All roots of the equation, $(1+z)^6 + z^6 = 0$:

(A) lie on a unit circle with centre at the origin

(B) lie on a unit circle with centre at (–1, 0)

(C) lie on the vertices of a regular polygon with centre at the origin

(D) are collinear

समीकरण $(1+z)^6 + z^6 = 0$ के सभी मूल :

(A) केन्द्र (0, 0) वाले इकाई वृत्त पर होंगे

(B) केन्द्र (- 1, 0) वाले इकाई वृत्त पर होंगे

(C) केन्द्र (0,0) वाले सम बहुभुज के शीर्षो पर होंगे

(D) संरेखीय है

Ans. D

Sol. $z = -\frac{1}{2}\left(1 + i\cot\frac{2r+1}{12}\pi\right)$, $r = 1, 2, 3, 4, 5$

7. It is given that complex numbers z_1 and z_2 satisfy $|z_1| = 2$ and $|z_2| = 3$. If the included angle of their corresponding vectors is 60° then $\left|\frac{z_1 + z_2}{z_1 - z_2}\right|$ can be expressed as $\frac{\sqrt{N}}{7}$ where N is natural number then N equals :

सम्मिश्र संख्याऐं z_1 और z_2, $|z_1| = 2$ और $|z_2| = 3$ को सन्तुष्ट करते हैं तथा इनके संगत सदिशों के मध्य का कोण 60° है तब $\left|\frac{z_1 + z_2}{z_1 - z_2}\right|$ को, $\frac{\sqrt{N}}{7}$ से दर्शाये तब N (प्राकृत संख्या) बराबर :

(A) 126 (B) 119 (C) 133 (D) 19

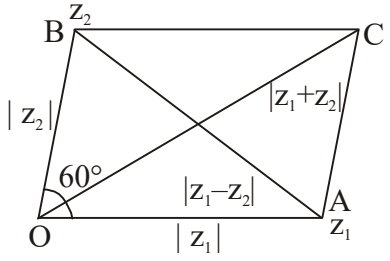
Ans. C

Sol. Using cosine rule

$$|z_1 + z_2| = \sqrt{|z_1|^2 + |z_2|^2 - 2|z_1||z_2|\cos 120°}$$

$$= \sqrt{4 + 9 + 2\cdot 3} = \sqrt{19}$$

and $|z_1 - z_2| = \sqrt{|z_1|^2 + |z_2|^2 - 2|z_1||z_1|\cos 60°}$



$$= \sqrt{4+9-6} = \sqrt{7}$$

$$\therefore \quad \left|\frac{z_1 + z_2}{z_1 + z_2}\right| = \sqrt{\frac{19}{7}} = \frac{\sqrt{133}}{7} \quad \Rightarrow N = 133$$

8. Let $f(x) = x^3 + ax^2 + bx + c$ be a cubic polynomial with real coefficients and all real roots. Also $|f(i)| = 1$ where $i = \sqrt{-1}$:

Statement-1 : All 3 roots of $f(x) = 0$ are zero

Statement-2 : $a + b + c = 0$

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false

(D) Statement-1 is false, statement-2 is true

माना $f(x) = x^3 + ax^2 + bx + c$ वास्तविक गुणांकों तथा वास्तविक मूलों का त्रिघातीय बहुपद है। इस प्रकार कि $|f(i)| = 1$ जहां $i = \sqrt{-1}$

कथन–1 : $f(x) = 0$ के तीनों मूल शून्य है।

कथन–2 : $a + b + c = 0$

(A) कथन-1 सत्य है, कथन-2 सत्य है ; कथन-2, कथन-1 का सही स्पष्टीकरण है

(B) कथन-1 सत्य है, कथन-2 सत्य है ; कथन-2, कथन-1 का सही स्पष्टीकरण नहीं है

(C) कथन-1 सत्य है, कथन-2 असत्य है

(D) कथन-1 असत्य है, कथन-2 सत्य है

Ans. B

Sol. Let $x_1, x_2, x_3 \in R$ be the roots of $f(x) = 0$

$\therefore \quad f(x) = (x - x_1)(x - x_2)(x - x_3)$

$f(i) = (i - x_1)(i - x_2)(i - x_3)$

$|f(i)| = |x_1 - i|\ |x_2 - i|\ |x_3 - i| = 1$

$\therefore \quad \sqrt{x_1^2 + 1}\ \sqrt{x_2^2 + 1}\sqrt{x_3^2 + 1} = 1$

This is possible only if $x_1 = x_2 = x_3 = 0$

$\Rightarrow \quad f(x) = x^3 \quad \Rightarrow \quad a = 0 = b = c \quad \Rightarrow \quad a + b + c = 0$

(sum of coeffiients zero can not imply that all zero roots)

9. If $z = \frac{\sqrt{3} - i}{2}$, then $(i^{101} + z^{101})^{103}$ equals :

(A) $\bar{z}$ (B) z (C) $\overline{z}$ (D) None of these

यदि $z = \frac{\sqrt{3} - i}{2}$, तब $(i^{101} + z^{101})^{103}$ बराबर है :

(A) $i\bar{z}$ (B) z (C) $\bar{z}$ (D) इनमें से कोई नहीं

Ans. B

Sol. We have, $z = \frac{\sqrt{3}-i}{2} = e^{i\left(-\frac{\pi}{6}\right)}$

$i^{101} = i^{4\times25+1} = i$

$z^{101} = e^{101i\left(-\frac{\pi}{6}\right)} = e^{-i(6\times17-1)\left(\frac{\pi}{6}\right)} = e^{-i\left(17-\frac{1}{6}\right)\pi}$

$z^{101} = e^{-i17\pi}e^{i\frac{\pi}{6}} = -1\times e^{i\left(\frac{\pi}{6}\right)} = -e^{i\left(\frac{\pi}{6}\right)}$

$\left(i^{101}+z^{101}\right)^{103} = \left(i - e^{i\left(\frac{\pi}{6}\right)}\right)^{103}$

$= \left(i - \frac{\sqrt{3}}{2} - \frac{i}{2}\right)^{103}$

$= \left(e^{5i\frac{\pi}{6}}\right)^{6\times17+1} = e^{(17\times5)i\pi}e^{5i\frac{\pi}{6}} = -1\times\left(\frac{-\sqrt{3}}{2}+\frac{i}{2}\right)$

10. $(-64)^{1/4}$ equals :

(A) $\pm 2(1+i)$ (B) $\pm 2(1-i)$ (C) $\pm 2(1\pm i)$ (D) None of these

$(-64)^{1/4}$ बराबर है :

(A) $\pm 2(1+i)$ (B) $\pm 2(1-i)$ (C) $\pm 2(1\pm i)$ (D) इनमें से कोई नहीं

Ans. C

Sol. Let $z = (-1)^{\frac{1}{4}}$

$z^4 = -1$

$z^4 = \cos n\pi + i \sin n\pi$

$z^4 = e^{n\pi}$

$z = e^{\frac{n\pi}{4}}$

Let n = 1

Now, $(-64)^{1/4}$

$\Rightarrow (-1)^{1/4} \cdot (64)^{1/4}$

$\Rightarrow E^{\frac{\pi}{4}}.\left(2^6\right)^{\frac{1}{4}}$

$\Rightarrow \frac{1+i}{\sqrt{2}}.2\sqrt{2} = 2(1+i)$

$\left(-64\right)^{\frac{1}{4}} = -2(1+i)$

$\Rightarrow \pm 2(1+i)$

11. If α is non real and $\alpha = \sqrt[5]{1}$ then the value of $2^{|1+\alpha+\alpha^2+\alpha^{-2}-\alpha^{-1}|}$ is equal to :

(A) 4 (B) 2 (C) 1 (D) None of these

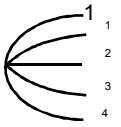
यदि α अवास्तविक हैं एवं $\alpha = \sqrt[5]{1}$ तो $2^{|1+\alpha+\alpha^2+\alpha^{-2}-\alpha^{-1}|}$ का मान हैं :

(A) 4 (B) 2 (C) 1 (D) इनमें से कोई नहीं

Ans. A

Sol. $\alpha = 1^{1/5}$

consider $x^5 - 1 = 0$

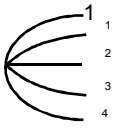


so $2^{\left|1+\alpha+\alpha^2+\frac{\alpha^3}{\alpha^5}-\frac{\alpha^4}{\alpha^5}\right|} = 2^{|1+\alpha+\alpha^2+\alpha^3-\alpha^4|}$

$= 2^{|-\alpha^4-\alpha^4|} = 4^{|\alpha^4|} = 4$

Hindi $\alpha = 1^{1/5}$

$x^5 - 1 = 0$ लेने पर



अतः $2^{\left|1+\alpha+\alpha^2+\frac{\alpha^3}{\alpha^5}-\frac{\alpha^4}{\alpha^5}\right|} = 2^{|1+\alpha+\alpha^2+\alpha^3-\alpha^4|}$

$= 2^{|-\alpha^4-\alpha^4|} = 4^{|\alpha^4|} = 4$

12. If $p = a + b\omega + c\omega^2$; $q = b + c\omega + a\omega^2$ and $r = c + a\omega + b\omega^2$ where $a, b, c \neq 0$ and ω is the imaginary cube root of unity, then :

(A) $p + q + r = a + b + c$ (B) $p^2 + q^2 + r^2 = a^2 + b^2 + c^2$

(C) $p^2 + q^2 + r^2 = -2(pq + qr + rp)$ (D) none of these

यदि $p = a + b\omega + c\omega^2$; $q = b + c\omega + a\omega^2$ एवं $r = c + a\omega + b\omega^2$, जहाँ a, b, c $\neq 0$ हो तथा इकाई का सम्मिश्र घनमूल ω हो, तो –

(A) $p + q + r = a + b + c$ (B) $p^2 + q^2 + r^2 = a^2 + b^2 + c^2$

(C) $p^2 + q^2 + r^2 = 2(pq + qr + rp)$ (D) इनमें से कोई नहीं

Ans. C

Sol. $p + q + r = (a + b + c)(1 + \omega + \omega^2) = 0$

$\Rightarrow \quad (p + q + r)^2 = 0 \quad \Rightarrow \quad p^2 + q^2 + r^2 + 2pq + 2qr + 2rp = 0$

& $\quad p^2 + q^2 + r^2 = (a^2 + b^2 + c^2)(1 + \omega + \omega^2) + 2(ab + bc + ca)(1 + \omega + \omega^2) = 0$

$\therefore pq + qr + rp = 0$

13. If $x^2 + x + 1 = 0$ then the numerical value of;

$$\left(x+\frac{1}{x}\right)^2+\left(x^2+\frac{1}{x^2}\right)^2+\left(x^3+\frac{1}{x^3}\right)^2+\left(x^4+\frac{1}{x^4}\right)^2+.....+\left(x^{27}+\frac{1}{x^{27}}\right)^2 =$$

(A) 54 (B) 36 (C) 27 (D) 18

यदि $x^2 + x + 1 = 0$ हो, तो $\left(x+\frac{1}{x}\right)^2+\left(x^2+\frac{1}{x^2}\right)^2+\left(x^3+\frac{1}{x^3}\right)^2+\left(x^4+\frac{1}{x^4}\right)^2+.....+\left(x^{27}+\frac{1}{x^{27}}\right)^2$ का संख्यात्मक मान है –

(A) 54 (B) 36 (C) 27 (D) 18

Ans. A

Sol. $\because$ roots are ω and ω^2

$$\left(\omega+\frac{1}{\omega}\right)^2+\left(\omega^2+\frac{1}{\omega^2}\right)^2+\left(\omega^3+\frac{1}{\omega^3}\right)^2.....\left(\omega^{27}+\frac{1}{\omega^{17}}\right)^2$$

there are 9 term which have ω^{3p}. so sum $9 \times 4 = 36$ there are 18 term which not have ω^{3p}

so sum is = 18. Total sum = 18 + 36 = 54

Hindi. चूंकि मूल ω और ω^2 है।

$$\left(\omega+\frac{1}{\omega}\right)^2+\left(\omega^2+\frac{1}{\omega^2}\right)^2+\left(\omega^3+\frac{1}{\omega^3}\right)^2.....\left(\omega^{27}+\frac{1}{\omega^{17}}\right)^2$$

9 पद जिनमें ω^{3p} के पद है। इसलिए योगफल $9 \times 4 = 36$ यहाँ 18 पद जो ω^{3p} के नही है।

इसलिए योगफल = 18 कुल योगफल = 18 + 36 = 54

14. ω is an imaginary cube root of unity. If $(1 + \omega^2)^m = (1 + \omega^4)^m$, then least positive integral value of m is

(A) 6 (B) 5 (C) 4 (D) 3

ω इकाई का काल्पनिक घन मूल है। यदि $(1 + \omega^2)^m = (1 + \omega^4)^m$, तो m का न्यूनतम धनात्मक पूर्णांक मान हैं–

(A) 6 (B) 5 (C) 4 (D) 3

Ans. D

Sol. $(1 + \omega^2)^n = (1 + \omega^4)^n \Rightarrow (-\omega)^n = (-\omega^2)^n$ which is true for n = 3 for least positive integer

Hindi. $(1 + \omega^2)^n = (1 + \omega^4)^n \Rightarrow (-\omega)^n = (-\omega^2)^n$ जो कि न्यूनतम धनात्मक पूर्णांक n = 3 के लिए सत्य है।

15. If a, b, c are integers not all equal and w is a cube root of unity (w ≠ 1), then the minimum value of $|a + bw + cw^2|$ is

(A) 0 (B) 1 (C) $\sqrt{3}/2$ (D) 1/2

a, b, c पूर्णांक संख्याएँ है जो कि एक साथ बराबर नही है और इकाई का घनमूल w है (w ≠ 1), तब $|a + bw + cw^2|$ का न्यूनतम मान होगा।

(A) 0 (B) 1 (C) $\sqrt{3}/2$ (D) 1/2

Ans. B

Sol. Let माना $p = |a + b\omega + c\omega^2|$

$p^2 = (a + b\omega + c\omega^2)\ (a + b\bar{\omega} + c\bar{\omega}^2) = (a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$

$p^2 = (a^2 + b^2 + c^2 - ab - bc - ca)$

$p^2 = \frac{1}{2}((a-b)^2 + (b - c)^2 + (c - a)^2)$

Since a, b, c are all integers but not all equal simultaneously

चूंकि a, b, c ∈ I लेकिन एक साथ समान नहीं है।

so RHS will be minimum only when any two are zero and third is a minimum magnitude integer = 1

a = b = 0, c = 1

इसलिए RHS न्यूनतम तभी होगा जब दो शून्य हो व तीसरा न्यूनतम पूर्णांक 1 हो।

a = b = 0, c = 1

so $p^2 \geq \frac{1}{2}[0 + 1 + 1]$

$p^2 \geq 1$

minimum value of |p| = 1

|p| का न्यूनतम मान = 1